The Distributional Consequences of Climate Change: Housing Wealth, Expectations, and Uncertainty

[Click here for latest version]

Jeffrey E. Sun*

October 16, 2026

Abstract

The median US household holds most of its wealth in one immobile, illiquid asset: the home. How does this shape the distributional incidence of climate change? I investigate with a 2447-location quantitative heterogeneous-agent model, globally solved with a stochastic climate process. Households are forward-looking, mobile, and save in bonds and, optionally, housing. Climate expectations shape fully-endogenized, spatially-segmented housing prices and rents, reallocating $95.58bn across space by 2030 and $907.8bn by 2100. Climate uncertainty generates high uncertainty in housing prices but only after 30-40 years in the future. Most housing wealth reallocation is due more to reallocation of construction across space than to contemporaneous price impacts of climate surprises. Construction reallocation also reduces the right tail of disaster damage burdens across scenarios. Policy analysis suggests that a flood insurance subsidy is not welfare-improving overall, even for generations alive now.

JEL codes: Q54, R31, R11, R13, E21, D31, C63.

*University of Toronto. Email: je.sun@utoronto.ca.

I am deeply indebted to my advisors, Ezra Oberfield, Stephen Redding, Richard Rogerson, and Ishan Nath, for their thoughtfulness and generosity. This project was completely derailed (for the better) by conversations with Monika Mrázová, Dávid Nagy, Luigi Pascali, Gianluca Violante, and Motohiro Yogo. I am grateful for the detailed comments of Leah Boustan, Levi Crews, José-Luis Cruz, Mayara Felix, Allan Hsiao, Ricardo Lagos, Lukas Mann, and John Sturm Becko. I have greatly benefited from conversations with Elena Aguilar, David Argente, Marnie Ginis, John Grigsby, Gene Grossman, Sebastián Guarda, Jacob Hartwig, Michael Jenuwine, Nobuhiro Kiyotaki, Benny Kleinman, Hugo Lhuillier, Mikhail Mamonov, Kristina Manysheva, Pablo Ottonello, Esteban Rossi-Hansberg, Anna Pestova, Thomas Sargent, Karthik Sastry, William Toms, Robert Wagner, Mark Watson, Sifan Xue, Yucheng Yang, and many seminar participants. I thank Zhongdao Wang for excellent research assistance. I gratefully acknowledge The Simpson Center for the Study of Macroeconomics and the International Economics Section at Princeton for their generous funding. Acknowledgements owed to the providers of the data this paper uses, and the terms of use governing them, are recorded in Appendix B.6; the software the quantification runs on is credited in Appendix D.4.

1 Introduction

The effects of climate change are highly uneven across space, but how does that translate into heterogeneous impacts at the household level? In a world in which people could move costlessly, a household's current location would be largely irrelevant to the climate impact they suffer, but we do not live in such a world. In a world in which households could avoid climate impacts simply by moving, policies which make it easier for households to move would reduce households' exposure to local climate impacts. But when U.S. homeowners are, on average, highly exposed to the prices of their homes, and those prices are climate sensitive, welfare impacts occur through the revaluation of immobile housing wealth and cannot be avoided by moving. Nor are they necessarily attenuated by policies which ease moving.

A complexity of the housing wealth channel of climate impacts is that housing prices reflect expectations about future conditions, rather than current conditions in the moment. Furthermore, upward-sloping housing supply curves cause housing *quantities*, in addition to prices, to respond to climate expectations and to uncertainty in climate expectations as well.

I study this housing wealth channel of climate impacts through the lens of a 2447-location spatial model of the contiguous U.S. The model explicitly incorporates overlapping generations of forward-looking heterogeneous households. Households differ in wealth and income and make forward-looking decisions about migration, homeownership, and savings in both liquid assets and real estate, under both idiosyncratic income and aggregate climate risk. Real estate prices, quantities, and rents in each location are jointly endogenously determined through the interaction of households on both sides of spatially-segmented real estate and housing rental markets. The model is shocked by a climate process which impacts amenities, productivity, and disaster damage exposure unevenly across space. The aggregate climate process is stochastic. Climate information is revealed, and climate uncertainty resolved, gradually over time. The model simulates 200 years of stochastic climate change, starting at a 2020 initial state, and is solved globally. This model is, to my knowledge, the first globally-solved heterogeneous-agent model with aggregate risk, many locations, and a household balance sheet.

In the calibrated model, the estimated effect of climate expectations on housing wealth is a reallocation across space of $95.58bn in 2030 and $907.8bn by 2100. Climate uncertainty generates high uncertainty in housing prices but only after 30-40 years in the future. Most housing wealth reallocation is due more to reallocation of construction across space than to contemporaneous price impacts of climate surprises. Construction reallocation also reduces the right tail of disaster damage burdens across scenarios. Policy analysis suggests that a flood insurance subsidy is not welfare-improving overall, even for generations alive now. A simple managed-retreat policy is considered, which expands housing supply in locations close to flood zones, and it found to be non-welfare-improving, above the baseline benefits of housing supply expansion.

Solving the model globally is challenging but necessary for this application. It allows us to examine the consequences of different aggregate climate paths drawn from a distribution that households are aware of ex ante. It also produces the endogenous prices responses to uncertainty

which would be absent in the perfect-foresight case or a first-order perturbation solution. The problem is intractable with existing conventional or deep-learning-based methods, and is solved by the highly-general method of neural Value Function Iteration (nVFI), which was developed for this application and presented in full generality in Sun (2026a) and Sun (2026b).

Quantifying the model requires four separate but connected efforts. First, matching aggregate and distributional features of the baseline U.S. economy. Second, matching spatial features of the U.S. economy location-by-location. Third, quantifying heterogeneous amenity, productivity, and disaster damage exposures to global climate change, location-by-location. Fourth, quantifying the stochastic global climate process, which involves not only the anticipated paths, but the variance and persistence of that process: how uncertain are aggregate climate expectations, and how much do many-decades-ahead climate expectations respond to information revealed in the present?

To perform the quantification, I use an internal calibration approach for the first set of parameters. This nests a spatial inversion loop adapted from the quantitative spatial literature which quantifies the second set of parameters. The third set of parameters is quantified by combining spatial climate projections with additive-separable damage functions in the manner of recent spatial climate impacts literature. For the fourth set, I use global climate projections along the representative concentration pathways to set the shape of different climate paths. I find little data to discipline the variance or persistence of a process which nests these paths, and therefore report results for a range of variances and persistences.

Combining these quantification methods into an internally consistent overall quantification requires a substantial effort with numerous subtle challenges. I therefore consider this joint model quantification to be part of this paper's contribution.

The mechanisms driving the results in this paper are individually widely-accepted and well-documented: climate change affects different locations differently, spatial shocks affect housing prices, households are exposed to those prices, in equilibrium mobility can spread out spatial shocks for mobile factors and concentrate them for fixed factors, prices are based on expectations, and uncertainty dampens investment. This paper builds on the literatures that establish and explore these mechanisms individually or in partial combination. The execution and coherent integration of these in a single model is where this paper breaks new ground.

The spatial incidence of climate damages is a rich area of active research. Hsiang et al. (2017) estimate and harmonize highly spatially-uneven climate impacts on disasters, mortality, productivity, energy costs, and crime. Carleton et al. (2022) focus on mortality and accounting for the U-shaped relationship between temperature and mortality and estimating the costs and benefits of adaptation. Rode et al. (2021) focus on spatial heterogeneity in the effects of climate change on energy consumption.

A large literature studies the housing price and household wealth impacts of shocks, with and without space, through the household's balance sheet. Yao and Zhang (2005) and Sinai and Souleles (2005) study the household's decision problem under housing market risk. Greaney (2023) and Giannone et al. (2023) build quantitative spatial models with homeownership and wealth

in continuous-time and discrete-time, respectively. Favilukis, Ludvigson, and Van Nieuwerburgh (2017) and Kaplan, Mitman, and Violante (2020) study the impact and household incidence of housing market shocks and uncertainty in equilibrium, in the presence of household life cycle dynamics and credit constraints.

The study of the role of uncertainty in shaping climate impacts is advanced by Cai and Lontzek (2019), and the importance of climate uncertainty for asset pricing generally is studied seminally by Bansal, Kiku, and Ochoa (2016). Barnett, Brock, and Hansen (2020) further study the decomposition of uncertainty into risk, ambiguity, and misspecification, outside of an equilibrium setting. Climate uncertainty is a rich topic, and papers such as Lemoine and Traeger (2014) on tipping points and Golosov et al. (2014) on optimal mitigation policy in the presence of technological and emissions uncertainty go further than this paper in disentangling the features and implications of climate uncertainty. Pang and Sun (2023) study the U.S. disaster relief policies in a globally-solved spatial model with forward-looking, heterogeneous households, but do not incorporate household liquid or housing wealth.

The effect of climate expectations on housing prices is studied by Bernstein, Gustafson, and Lewis (2019), Murfin and Spiegel (2020), and Bakkensen and Barrage (2022) in reduced form. Hsiao (2023) goes further in studying how imperfect policy commitment can affect anticipatory household responses and limit the scope of managed retreat. Balboni (2021) studies optimal policy and climate impacts in spatial equilibrium quantitatively, in a study similar in spirit to this paper, but focusing on regional trade rather than housing markets.

This paper is, to my knowledge, the first to globally solve a heterogeneous-agent model with aggregate risk, many locations, and a household balance sheet. On the climate side, Bilal and Rossi-Hansberg (2023) break new ground by studying how anticipatory migration shapes climate impacts. Their model features immobile productive capital, which mine does not, but it is owned by immobile capitalists rather than their mobile households who are hand-to-mouth. Cruz and Rossi-Hansberg (2024) and Desmet et al. (2021) enrich the set of adaptation margins that can be tractably incorporated in a model, including trade, natality, and endogenous technology, which I do not, but feature no household heterogeneity. None incorporate housing wealth. Fried (2022) does feature forward-looking heterogeneous households who can save, choose to own housing, and engage in two-sided housing markets, but the solution is a comparative statics exercise and space is limited to two locations. Krusell and Smith (2022) find a strong role of capital mobility in shaping adaptation in spatial equilibrium, in a model in which labor is immobile.

Without climate change, there is a burgeoning literature on how household balance sheets, lifecycle dynamics, and frictions affect spatial decisions and equilibria. These include Crews (2023) introducing endogenous human capital accumulation, Komissarova (2022) introducing the interaction of spatial equilibrium and retirement lifecycle dynamics, and Kleinman, Liu, and Redding (2023) introducing endogenous dynamic fixed capital accumulation in a spatial equilibrium setting admitting exact-hat algebra.

In computing the global solution, the combination of household heterogeneity and aggregate

uncertainty rule out conventional methods. High-resolution space rules out perceived-law-of-motion based on Krusell and Smith (1998). Occasionally-binding constraints, housing illiquidity, discrete household migration decisions and the spatial nature of the aggregate state space rule out existing deep-learning methods. The household problem of this paper mixes multiple discrete and continuous actions, with discontinuous policies over continuous actions, which does not admit a trainable neural network representation using the methods of existing deep-learning based approaches such as Han, Yang, and E (2021) and Azinovic, Gaegauf, and Scheidegger (2022). Of these, Gu et al. (2024) and Fernández-Villaverde et al. (2020) are able to accommodate space, but not the full household problem.

To overcome this, nVFI uses conventional methods to solve for household policies and prices. This approach is related to the parameterized-expectations approaches of den Haan and Marcet (1990) and Maliar, Maliar, and Winant (2021). nVFI is highly general but requires a large amount of training data of a kind which can be produced by conventional methods, but is computationally expensive without extremely optimized code in a model with a five-dimensional household state space with 66.29 million gridpoints. To make nVFI feasible, a highly general method for solving heterogeneous-agent household blocks was developed (Sun 2026b).

The paper is structured as follows. In Section 2, I describe the model. In Section 3, I describe the quantification of the spatial economy. In Section 4, I describe the construction of the mapping from global climate to local damages. In Section 5, I describe the global climate process and the range of parameterizations used for the results. In Section 6, I present results on the quantified demographic and spatial incidence of climate change, and the regressive distributional consequences of climate uncertainty. In Section 7, I analyze the effects of land-use deregulation, flood-insurance subsidization, migration-easing policy, and improvements in long-run climate forecasts.

2 A Dynamic Spatial Equilibrium Model

This section develops a discrete-time, heterogeneous-agent, dynamic spatial model of the United States housing market under climate uncertainty. The economy consists of a finite set of locations $\ell \in \mathcal{L}$, representing the contiguous United States as a small open economy. It is populated by overlapping generations of forward-looking households who make location, saving, consumption, and real estate investment decisions subject to spatial and financial frictions and to both idiosyncratic and aggregate risk.

Households may save in a risk-free foreign-issued bond at a fixed exogenous interest rate or invest in residential real estate. Real estate is illiquid, indivisible, and risky: housing prices and rents respond endogenously to climate shocks, and this risk is neither insurable by households nor diversifiable across locations. Households financing real estate purchases may borrow against owned housing subject to a collateral constraint, paying a mortgage rate strictly higher than the risk-free rate. Transacting in real estate incurs a proportional fee, and housing may only be owned in the location in which the household resides.

Housing supply in each location consists of a homogeneous stock of durable housing that depreciates each period and is augmented by a competitive construction sector with a location-specific, upward-sloping supply curve. Because markets are spatially segmented and ownership is tied to residence, equilibrium housing prices and rents are determined location by location.

A government operates a subsidized national flood insurance program, fixing premia at a constant discount to the actuarially fair level. I abstract from realized idiosyncratic disaster damages: the insurer pays to restore all damages within the period. Households are therefore insured against realized disasters but remain exposed to changes in disaster *risk*, which pass through to insurance premia.

Climate change introduces aggregate uncertainty. A stochastic global temperature process shifts local labor productivity, amenities, and disaster intensity. Climate elasticities are heterogeneous across space but constant within each location. Households are exposed both to these direct climate impacts and to the endogenous response of housing prices and rents that they imply in equilibrium.

2.1 Households

2.1.1 Life Cycle

Households live for $a_{\max} = 6$ periods, each representing one decade, spanning ages 20–29 through 70–79. Households are indexed by $i \in I$, calendar decades by t, and age by $a_{it} \in \{1, \ldots, a_{\max}\}$. A new cohort of age-one households enters each period, and households who reach the end of their sixth period die and derive warm-glow bequest utility from their liquidated assets.

2.1.2 Preferences

Each period, a household i living in location ℓ derives flow utility from non-housing goods g_{it}, housing services h_{it}^{rent}, and location-specific amenities $\alpha_{\ell t}$. Consumption utility is CES between goods and housing services with elasticity of substitution σ, with constant elasticity of intertemporal substitution $1/\eta$:

$$u(g, h^{\text{rent}}, \alpha) = \frac{\left[\alpha \left((1-\gamma) g^{\frac{\sigma-1}{\sigma}} + \gamma h^{\text{rent}\,\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}}\right]^{1-\eta} - 1}{1-\eta}.$$

Location-specific amenities $\alpha_{\ell t}$ enter as a multiplier on the consumption aggregate and are directly affected by the climate state.

2.1.3 Migration

Each period, households decide whether to stay in their current location or move to a new location. To move, households must sell their home. After selling their home, before making the destination

decision, each household draws i.i.d. additive location preference shocks,

$$\varepsilon_{\ell it} \sim \text{Gumbel}\left(0, \psi^{-1}\right),$$

one for each candidate location $\ell \in \mathcal{L}$. The scale parameter ψ^{-1} governs the dispersion of idiosyncratic tastes for locations.

Migrating from ℓ to ℓ' incurs a bilateral utility cost $F(\ell, \ell') \geq 0$ with $F(\ell, \ell) = 0$.

2.1.4 Endowments

Each household i inelastically supplies an effective labor endowment z_{it} in its current location of residence. At time t, conditional on living in location ℓ, the household's earnings are $A_{\ell t} z_{it}$, where $A_{\ell t}$ is a location- and time-specific productivity that is directly affected by the climate state. The idiosyncratic component z_{it} evolves according to an age-dependent log-AR(1) process,

$$\begin{aligned} \log z_{it} &= u_{it} + \zeta_{a_{it}} \\ u_{it+1} &= p u_{it} + \nu_{it} \\ \nu_{it} &\sim \mathcal{N}(0, \sigma_\nu^2) \quad \text{i.i.d.}, \end{aligned}$$

with persistence $p \in [0, 1]$ and age-specific shifters ζ_a.[1]

2.1.5 Assets

Households hold two classes of assets. The first is a liquid, risk-free bond b_{it} issued overseas, with exogenous net return r^f. Bondholdings may be negative, in which case they represent mortgage debt. The second is residential real estate in the household's current location h_{it}^{own}. Real estate is indivisible and subject to transaction costs. A household moving across locations must sell all of its real estate before moving.

2.1.6 Residential Real Estate

Real estate in location ℓ at date t trades at price $q_{\ell t}$, and rents at price $\rho_{\ell t}$, both determined in local equilibrium. All households rent housing; those who own real estate can be thought of as renting to themselves.[2] Renters pay $\rho_{\ell t}$ per unit of housing services consumed, while owners receive

$$\rho_{\ell t}^{\text{net}} = (1 - \chi)\rho_{\ell t} - \delta_{\ell t}^{\text{insurance}} \tag{1}$$

per unit of real estate owned. Maintenance costs χ generate a realistic wedge between housing prices and rents. $\delta_{\ell t}^{\text{insurance}}$ is a mandatory flood insurance premium.

1. This is isomorphic to the idiosyncratic human capital process of Kaplan, Mitman, and Violante (2020), except that the aggregate productivity shifter $A_{\ell t}$ is factored out of the endowment process and made location-specific.

2. While households can choose to own zero real estate, there is no explicit rent vs. own decision as in Kaplan, Mitman, and Violante (2020) or Giannone et al. (2023).

Insurance All properties are automatically enrolled in a subsidized National Flood Insurance Program operated by the government (Section 2.3), which charges a mandatory location-specific premium per unit of housing $\delta_{\ell t}^{\text{insurance}}$ to the owner. Each period, the insurer pays to exactly restore any realized disaster damages, so that the physical housing stock is preserved period-to-period. Households therefore face no idiosyncratic damage risk, but remain exposed to climate risk through the premium, which responds to climate-driven changes in disaster risk.

Real Estate Illiquidity Any adjustment to real estate holdings incurs a realtor's fee ϕ proportional to the sale price, evaluated at the *current* price in the *original* location[3] $q_{\ell_{i,t-1},t}$. After real estate adjustments, but before income is received and consumption expenses paid, bondholdings are then,

$$\tilde{b}_{it} = b_{it} + \left(1 - \mathbb{1}_{it}^{\text{sell}}\phi\right) q_{\ell_{i,t-1},t}\, h_{i,t-1}^{\text{own}} - q_{\ell_{it},t}\, h_{it}^{\text{own}}.$$

Mortgages Mortgages are interest-only and take the form of negative bondholdings. The interest rate charged on negative balances, r^m, strictly exceeds the return on positive balances, r^f:

$$r^m > r^f.$$

Mortgages must be collateralized by owned housing, subject to a minimum equity share[4] $\kappa \in (0, 1)$. This loan-to-value constraint is enforced at two points in time: when households choose whether to sell and when households choose consumption and savings. When choosing whether to sell, households must sell if they violate,

$$b_{it} \geq -(1-\kappa)\, q_{\ell_{i,t-1},t}\, h_{i,t-1}^{\text{own}}.$$

When choosing consumption and savings, households must satisfy,[5]

$$b_{i,t+1} \geq -(1-\kappa)\, q_{\ell t} h_{it}^{\text{own}}.$$

2.1.7 Entry

Each period, a new cohort of age-one households enter. Entrants draw i.i.d. location preference shocks $\varepsilon_{\ell it}$,

$$\varepsilon_{\ell it}^{\text{enter}} \sim \text{Gumbel}\left(0, \psi^{-1}\right),$$

from the same distribution as the within-period location preference shocks, and choose their initial location ℓ_{it} to maximize expected lifetime utility. Entrants arrive with no housing assets, and draw

3. Because a household may only own housing in its current location, migration forces the sale of all owned housing.
4. Or maximum loan-to-value ratio $1 - \kappa$.
5. The constraint is enforced at the end of the period, after the consumption-savings choice is made.

initial bondholdings and labor endowment from fixed, independent distributions,

$$
\begin{aligned}
x_{it} &= (b_{it}, z_{it}, 0, \ell_{i,t-1}), \\
(b_{it} \mid a_{it} = 1) &\sim \lambda_b^{\text{init}} \\
(z_{it} \mid a_{it} = 1) &\sim \lambda_z^{\text{init}}.
\end{aligned}
$$

2.1.8 Bequest Utility

A household who has reached the maximum age $a_{it} = a_{\max}$ derives warm-glow bequest utility from the liquidation value of its assets at the end of the period, deflated by the local price index.[6] Formally,

$$
\begin{aligned}
V^{\text{end}}_{a_{\max}}(b, z, h^{\text{own}}, \ell) &= Q\,\frac{u(\tilde{b}^{\text{liq}}, \rho_{\ell t})^{1-\eta}}{1-\eta}, \\
\text{where} \quad \tilde{b}^{\text{liq}} &= b + (1-\phi)\, q_{\ell t}\, h^{\text{own}}, \\
\text{and} \quad u(\tilde{b}^{\text{liq}}, \rho_{\ell t}) &= \tilde{b}^{\text{liq}} \Big/ \left[\left((1-\gamma)^{\sigma} + \gamma^{\sigma} \rho_{\ell t}^{1-\sigma} \right)^{\frac{1}{1-\sigma}} \right],
\end{aligned}
$$

where the strength of the bequest motive is governed by the parameter $Q > 0$.

2.1.9 Timing

Within each period t, events unfold in the following 9 stages:

1. Housing prices $q_{\ell t}$ and rents $\rho_{\ell t}$ are observed. Housing stocks $\{H_{\ell t}\}_{\ell}$ depreciate at a rate δ^H in each location and are augmented by new construction.

2. Household portfolio values are recalculated based on current prices.

3. Households either sell all of their real estate $h^{\text{own}}_{i,t-1}$, incurring a realtor's fee, or none of it, subject to taste shocks $\varepsilon^h_{it}\big(\mathbb{1}^{\text{sell}}\big)$.

4. Renters and sellers draw location preference shocks $\varepsilon_{\ell it}$ and choose their new location $\ell \equiv \ell_{it}$.

5. Renters and sellers draw real estate taste shocks $\varepsilon^h_{ith'} \sim \text{Gumbel}(0, \xi^{-1})\ \forall\ h'$ i.i.d. and choose their new real estate holdings h^{own}_{it}.

6. Households receive income from labor, bond interest, and rental properties, and pay for maintenance, flood insurance and mortgage interest.

$$
\text{Net Income}_{it} = \underbrace{A_{\ell t} z_{it}}_{\text{labor}} + \underbrace{\Big(r^f \mathbb{1}[\tilde{b}_{it} \geq 0] + r^m \mathbb{1}[\tilde{b}_{it} < 0] \Big) \tilde{b}_{it}}_{\text{bond interest}} + \underbrace{\left((1-\chi)\rho_{\ell t} - \delta^{\text{insurance}}_{\ell t} \right) h^{\text{own}}_{it}}_{\text{net rental income}}
$$

6. The model does not consider the general-equilibrium implications of bequests: there is no explicit intergenerational wealth transmission, and a reduction in end-of-life savings does not feed back into the initial wealth of new entrants.

7. Households choose and pay for their consumption of goods g_{it} and housing services h_{it}^{rent}. This determines their end-of-period bondholdings $b_{i,t+1}$.

$$b_{i,t+1} = \tilde{b}_{it} + \text{Net Income}_{it} - g_{it} - \rho_{\ell t}\, h_{it}^{\text{rent}}$$

8. Households draw next-period labor endowment $z_{i,t+1}$.

9. Aggregate shocks, if any, are realized. Households age and time passes by one period.

Household State Variables and Value Functions To fully describe and numerically solve this model, it was necessary to define intermediate value functions and household state distributions for each of the 9 stages described above. I defer this full treatment to Appendix A, and describe only a subset here.

Denote the state of household i at the beginning and end of period t, respectively, by

$$x_{it}^{\text{start}} = (b_{it}, z_{it}, h_{i,t-1}^{\text{own}}, \ell_{i,t-1}) \text{ and } x_{it}^{\text{end}} = (b_{i,t+1}, z_{i,t+1}, h_{it}^{\text{own}}, \ell_{it}),$$

and denote its components by subscripts, i.e., $(x_{it}^{\text{start}})_z \equiv z_{it}$. Denote the distribution of household states x_{it}^{start} and x_{it}^{end} at the beginning and end of period t, respectively, by

$$\lambda_t^{\text{start}} \text{ and } \lambda_t^{\text{end}}.$$

We write the value function of a household of age a at the beginning of period t with state x_{it}^{start}, given the *beginning-of-period* aggregate state[7] Γ_t, as,

$$V_a^{\text{start}}(x_{it}^{\text{start}}; \Gamma_t).$$

In fact, as no aggregate shocks occur before Stage 9, all intermediate value functions before Stage 9 can be written as a function of idiosyncratic state and the beginning-of-period aggregate state Γ_t. For example, we write the value function at the end of Stage 7, after the household has made its consumption-savings decision,

$$V_a^{\text{postconsume}}(x_{it}^{\text{postconsume}}; \Gamma_t),$$

and the value function at the end of Stage 5, after the household has made its real estate decision,

$$V_a^{\text{postbuy}}(x_{it}^{\text{postbuy}}; \Gamma_t),$$

where x^{postbuy}, $x^{\text{postincome}}$ are the household's state at the end of stage 5 and 6, respectively.

We write the value function of a household of age a with state x_{it}^{end}, at the end of Stage 8 of

7. Γ_t includes λ_t^{start}. See Equation (6) below.

period t, as

$$V_a^{\text{end}}(x_{it}^{\text{end}}; \Gamma_t).$$

We call this the "end-of-period" value function, or "continuation value."[8]

2.1.10 The Household's Dynamic Optimization Problem

Households discount the future at rate β and maximize expected lifetime utility. The household's full period utility is,

$$u(g_{it}, h_{it}^{\text{rent}}, \alpha_{\ell t}) - F(\ell_{i,t-1}, \ell_{it}) + \varepsilon_{\ell it} + \varepsilon_{it}^{h}(\mathbb{1}_{it}^{\text{sell}}) + \varepsilon_{ith_{it}^{\text{own}}}^{h}.$$

The full problem in recursive form is described in Appendix A. To illustrate the setup, I include one equation of the problem here: in Stage 7, a household i in location ℓ solves,[9]

$$\begin{aligned} & \max_{g_{it}, h_{it}^{\text{rent}}} u(g_{it}, h_{it}^{\text{rent}}, \alpha_{\ell t}) + V_{a_{it}}^{\text{postconsume}}((b_{i,t+1}, z_{i,t}, h_{i,t}^{\text{own}}, \ell_{i,t}); \Gamma_t) \\ \text{s.t.} \quad & b_{i,t+1} \geq -(1-\kappa)\, q_{\ell,t-1} h_{it}^{\text{own}} \\ \text{where} \quad & b_{i,t+1} = \tilde{b}_{it} + \text{Net Income}_{it} - g_{it} - \rho_{\ell t}\, h_{it}^{\text{rent}}. \end{aligned}$$

2.2 Firms

2.2.1 Final Goods

Each location ℓ contains a representative final-goods firm which produces the costlessly tradable numeraire consumption good with a linear, labor-only technology and location-specific productivity $A_{\ell t}$:

$$\begin{aligned} & Y_{\ell t} = A_{\ell t}\, Z_{\ell t}, \\ \text{where} \quad & Z_{\ell t} = \int_{x^{\text{postbuy}} \in \mathbf{X}} x_z^{\text{postbuy}}\, \mathbb{1}\left[x_\ell^{\text{postbuy}} = \ell\right] d\lambda_t(x^{\text{postbuy}}), \end{aligned}$$

is aggregate local effective labor. Firms pay each worker their marginal product $A_{\ell t} z_{it}$.

2.2.2 Construction

In each location ℓ, a competitive construction sector produces new housing using a goods-only technology with external decreasing returns to scale. Observing the current real estate price $q_{\ell t}$ and the depreciated stock of existing housing $(1-\delta^H)H_{\ell,t-1}$ the construction sector builds $H_{\ell t} -$

8. It would be redundant to define a value function at the end of Stage 9, as this would exactly equal the value function at the beginning of Stage 1 of the following period. Intuitively, Stage 9 can be thought of as occurring *between* the end of period t and the beginning of period $t+1$.

9. The loan-to-value constraint is based on house price appraisal in the previous period, $q_{\ell,t-1}$. This improves numerical stability and means that households are not forced into forbidden states by price fluctuations before they have the ability to sell.

$(1-\delta^H)H_{\ell,t-1} \geq 0$ units of new housing at an upward-sloping, location-specific marginal cost, following[10] Greaney (2023):

$$c_\ell(H_{\ell t}) = (H_{\ell t}/\Pi_\ell)^{1/\theta_\ell}. \tag{2}$$

Because decreasing returns to scale are external, this cost applies to all new housing units. Competition implies that, whenever construction is active, $q_{\ell t} = c_\ell(H_{\ell t})$. When the current real estate price is insufficient to incentivize new construction, supply is pinned at $(1-\delta^H)H_{\ell,t-1}$,

$$H^S_{\ell t} = \begin{cases} c_\ell^{-1}(q_{\ell t}) & q_{\ell t} \geq c_\ell((1-\delta^H)\bar{H}_{\ell,t-1}), \\ (1-\delta^H)\bar{H}_{\ell,t-1} & q_{\ell t} < c_\ell((1-\delta^H)\bar{H}_{\ell,t-1}). \end{cases} \tag{3}$$

2.2.3 Banking

A competitive foreign banking sector intermediates liquid savings and mortgages. It accepts deposits at the exogenous rate r^f and extends mortgage loans at the exogenous rate $r^m > r^f$, subject to each household's collateral constraint.

2.3 Government

The government administers a subsidized National Flood Insurance program. For every insured property, the government pays a constant share $s^{\text{NFIP}} \in [0,1]$ of the actuarially-fair premium, so that the effective premium passed through to households is

$$\delta^{\text{insurance}}_{\ell t} = (1 - s^{\text{NFIP}})\, \delta^{\text{disaster}}_{\ell t},$$

The government finances the subsidy lump-sum from funds that would otherwise be used for public services that benefit all households equally and do not affect their decisions. I therefore omit public services from the model equations and welfare calculations, instead choosing to report welfare benefits of policies alongside their implied opportunity costs in the results section,

$$\text{Policy Cost}_t = \sum_\ell s^{\text{NFIP}} \delta^{\text{disaster}}_{\ell t} H_{\ell t}.$$

2.4 Housing and Real Estate Equilibrium

Local real estate and rental markets clear location by location.

Residential real estate. Aggregate demand for real estate ownership in location ℓ is,

$$H^D_{\ell t} = \int_{x^{\text{postbuy}} \in \mathbf{X}} x^{\text{postbuy}}_{h^{\text{own}}} \, \mathbb{1}\left[x^{\text{postbuy}}_\ell = \ell\right] d\lambda_t(x^{\text{postbuy}}).$$

10. While Greaney (2023) uses the housing supply elasticities of Saiz (2010), I use those of Baum-Snow and Han (2024).

The real estate market clearing condition is,

$$H_{\ell t}^D \leq H_{\ell t}^S, q_{\ell t} \geq 0, \left(H_{\ell t}^S - H_{\ell t}^D\right) q_{\ell t} = 0 \tag{4}$$

for all locations $\ell \in \mathcal{L}$. This is a complementarity condition. When disaster damages are sufficiently high, insurance premia can exceed rents, so that $q_{\ell t} = 0$ and $H_{\ell t}^D < H_{\ell t}^S$. Housing is free disposal in this case: $H_{\ell t} = H_{\ell t}^D$, and in particular the stock carried over to the following period is $H_{\ell t}^D$ rather than $H_{\ell t}^S$.

Rental housing. Aggregate demand for rental services is,

$$H_{\ell t}^{\text{rent}} = \int_{x \in \mathbf{X}} h^{\text{rent}}(x^{\text{postincome}}; \Gamma_t) \, \mathbb{1}\left[x_\ell^{\text{postincome}} = \ell\right] d\lambda_t(x).$$

The rental market clearing condition is,

$$H_{\ell t}^{\text{rent}} = \bar{H}_{\ell t}, \tag{5}$$

for all locations $\ell \in \mathcal{L}$.[11]

2.5 Climate Sensitivities

I adopt an elegant, novel strategy for capturing the complex and heterogeneous impact of global climate change on local economic fundamentals across space. Each model location's amenity value, productivity, and disaster intensity are allowed to depend on the global climate state in a way that is fully flexible across locations and impacted fundamentals, but parsimonious within each location-fundamental pair:

$$\begin{aligned}
\log \alpha_{\ell t} &= \log \bar{\alpha}_\ell + \alpha_\ell^g \, \text{GSAT}_t && \text{(Amenity)} \\
\log A_{\ell t} &= \log \bar{A}_\ell + A_\ell^g \, \text{GSAT}_t && \text{(Productivity)} \\
\delta_{\ell t}^{\text{disaster}} &= \bar{\delta}_\ell + \delta_\ell^g \, \text{GSAT}_t && \text{(Disaster Intensity)}
\end{aligned}$$

where GSAT_t is global-mean surface air temperature (GSAT) in period t, measured in °C as an anomaly relative to the baseline.

In Section 4, I demonstrate a procedure for combining rich global-to-local ensemble climate models and local damage functions to estimate these flexible climate sensitivities $\{A_\ell^g, \alpha_\ell^g, \delta_\ell^g\}_\ell$. I also provide evidence that the parsimonious within-location-fundamental specifications are reasonable over the range of global temperatures of interest.[12]

11. To precisely understand the sense in which there exist supply and demand curves which intersect at these prices $q_{\ell t}$ and $\rho_{\ell t}$, consider the model in "partial intertemporal equilibrium," shutting down intertemporal feedback effects of current prices on V^{end} and Γ_t. Consider a one-period model taking initial state Γ_t and terminal payoffs $V^{\text{end}}(x^{\text{end}})$ as given. Then the household's problem reduces to a well-defined static optimization problem, and aggregate demand for housing services and real estate ownership are well-defined functions of prices $q_{\ell t}$ and $\rho_{\ell t}$. See Appendix A.2.

12. They arise from the composition of two approximately linear relationships: (1) the correlation of each location's

The variable GSAT_t is normalized to zero in the baseline steady state of the model so that, at baseline, $A_\ell = \bar{A}_\ell$, $\alpha_\ell = \bar{\alpha}_\ell$, and $\delta_\ell^{\text{disaster}} = \bar{\delta}_\ell$ for all ℓ.

2.6 Aggregate State

The aggregate state at the beginning of Stage 1 of Period t, denoted Γ_t, consists of the household state distribution λ_t^{start}, carried-over housing stocks $\{H_{\ell t}\}_\ell$, and climate variables D_t,

$$\Gamma_t = (\lambda_t^{\text{start}}, \{H_{\ell,t-1}, q_{\ell,t-1}\}_\ell, D_t). \tag{6}$$

2.7 Steady State Equilibrium

A *stationary steady-state equilibrium* is a time-invariant aggregate state $\Gamma^* = (\lambda^{\text{start}*}, \{H_\ell^*\}_\ell, D^*)$ together with constant prices $\{q_\ell^*, \rho_\ell^*\}_\ell$ such that:

- household policy and value functions solve the recursive household problem (Equation (11) in Appendix A) given constant Γ^*;
- the beginning-of-period household distribution $\lambda^{\text{start}*}$ is stationary: the inflow of new entrants and the outflow of exiting households, combined with household policy functions, reproduce $\lambda^{\text{start}*}$ each period;
- housing stocks $\{H_\ell^*\}_\ell$ are constant: new construction exactly replaces depreciation in every location;
- prices q_ℓ^* and ρ_ℓ^* satisfy the market-clearing conditions (4) and (5) in every location.

2.8 Perfect Foresight Climate Transition

In a perfect-foresight climate transition, the global climate state D_t consists of a year and a global-mean surface air temperature,

$$D_t = (t, \text{GSAT}_t)$$
$$\text{where} \quad \text{GSAT}_t = \text{GSAT}(t).$$

The economy begins in a stationary steady state with GSAT $= 0$. Climate change begins unexpectedly at $t = 1$. Starting at $t = 2$, households correctly anticipate the entire future trajectory of GSAT_t.

A perfect-foresight transition path in dynamic equilibrium is a sequence of aggregate states $\Gamma_t = (\lambda_t^{\text{start}}, \{H_{\ell t}\}_\ell, D_t)$ and prices $\{q_{\ell t}, \rho_{\ell t}\}_\ell$ such that, for all $t \geq 1$:

- household policy and value functions solve the recursive household problem given Γ_t;

heating- and cooling-degree-days with global temperatures and (2) damage functions in these local temperature measures.

- the beginning-of-period household distribution λ_t^{start} is consistent with the previous period's distribution $\lambda_{t-1}^{\text{start}}$, the previous period's aggregate state Γ_{t-1}, and household policy functions;
- housing stocks $\{H_{\ell t}\}_\ell$ evolve according to the transition equation (3) given the previous period's held stocks $\{\bar{H}_{\ell,t-1}\}_\ell$ and current prices $\{q_{\ell t}\}_\ell$;
- prices $q_{\ell t}$ and $\rho_{\ell t}$ satisfy the market-clearing conditions (4) and (5) in every location.

I describe the full equilibrium definition and solution method in Appendix A.3.

2.9 Stochastic Climate Transition

In a stochastic climate transition, the global climate state D_t consists of a year, a climate scenario, and a global-mean surface air temperature,

$$\begin{aligned} D_t &= (t, s_t, \text{GSAT}_t) \\ \text{where} \quad \text{GSAT}_t &= \text{GSAT}(t, s_t) \\ \text{and} \quad s_t &\in \{1, \ldots, S\}. \end{aligned}$$

The economy begins in a stationary steady state with GSAT $= 0$. Climate change begins unexpectedly at $t = 1$ with $s_1 = s^{\text{init}}$. Starting at $t = 2$, households have rational expectations but not perfect foresight. At the end of each period, i.e. in Stage 9, the climate scenario s_t evolves according to a Markov process with time-invariant transition matrix $\mathcal{P}$,

$$s_{t+1} \sim \mathcal{P}(\cdot \mid s_t), \tag{7}$$

Note that the global temperature $\text{GSAT}_t = \text{GSAT}(t, s_t)$ itself does not follow a Markov process with time-invariant transition probabilities, as it depends explicitly on time t. Fixing $s_t = s_1$, the sequence $\{\text{GSAT}(t, s_1)\}_t$ can be thought of as a climate scenario, and is calibrated to the midpoint of RCP4.5 and RCP6.0.

Dropping time subscripts, the beginning-of-period aggregate state is

$$\Gamma = (\lambda^{\text{start}}, H, D).$$

A *recursive stochastic equilibrium* consists of:

- household policy and value functions solving the recursive household problem given Γ, as well as satisfying, for all Γ and x^{end},

$$V_a^{\text{end}}(x^{\text{end}}; \Gamma) = \sum_{s'=1}^{S} \mathcal{P}\big(s' \mid s\big) V_{a+1}^{\text{start}}(x^{\text{end}}; \Gamma'(\Gamma, s')) \tag{8}$$

- an aggregate transition function

$$\Gamma'(\Gamma, s')$$

- price functions $q_\ell(\Gamma)$ and $\rho_\ell(\Gamma)$ satisfying the market-clearing conditions (4) and (5) for every Γ, ℓ.

I describe the full equilibrium definition in Appendix A.4 and the solution method in Appendix D.

3 Baseline Calibration

The quantification of the model is implemented in four blocks: reproducing aggregate and distributional features of the U.S. economy in the 2020 baseline; matching the distribution of population, income, and housing across space; specifying how each location is exposed to global climate change and in which ways; and constructing a stochastic global climate process representing the shape of future anticipated climate paths, the level of ex ante uncertainty over those paths, and the rate at which that uncertainty is resolved.

These are described over the following three sections. This section, Section 3 describes the calibration of the initial steady state of the economy, at the aggregate and location level. Section 4 describes the construction of damage functions to latent parameters: how each location's productivities, amenities, and disaster damages are impacted by the global climate state. Section 5 describes the construction of the stochastic global climate process, representing ongoing uncertainty over climate scenarios and the ongoing resolution of that uncertainty. The full table of data sources appears in Appendix B.

3.1 Spatial and Non-Spatial Parameters

The initial steady state of the model is calibrated to match aggregate and distributional features of the U.S. economy in 2020, as well as features of each of 2447 locations. A subset of parameters is set externally because they are not separately identified by simulated moments. The rest are each associated with an informative simulated moment. Each location-level and non-spatial parameter is associated with a single moment at the location or economy-wide level, respectively. The full set of internally-calibrated parameters are exactly identified.

The global temperature variable GSAT_t is normalized to zero in the baseline steady state, so that, for example, where productivity has the functional form,

$$\log A_{\ell t} = \log \overline{A}_\ell + A^g_\ell \text{GSAT}_t,$$

and $\overline{A}_\ell$ is the baseline location-level productivity set in Section 3, A^g_ℓ is the location-level sensitivity of productivity to global temperature set in Section 4, and GSAT_t is zero initially and subsequently evolves according to the stochastic process constructed in Section 5.

3.2 Inner Loop: Spatial Parameters

The spatial unit is the 2020 Public Use Microdata Area (2020 PUMAs, henceforth "PUMAs"). These are defined by the U.S. Census Bureau and constructed by population, so that each has between 100,000 and 200,000 residents per PUMA in 2020. Conveniently, this causes PUMAs to have higher spatial resolution along coastlines where flood risk varies most granularly, while remaining coarse over large inland regions.

2020 housing stocks H_ℓ^S are taken from the Census Bureau. Baseline average annual losses from disaster damages $\overline{\delta}_\ell$ are restricted to flooding and taken from Bates et al. (2021). The raw damage estimates are given for all structures, and I uniformly deflate them by the aggregate share of residential real estate in total structures, 0.655 in First Street Foundation (2021).

Housing supply elasticities θ_ℓ are taken from Baum-Snow and Han (2024) subject to imputations and a crosswalk procedure described in Appendix B.[13]

The *inner loop* of the internal calibration procedure is as follows. Conditional on a particular guess of non-spatial parameters, location-level amenities $\overline{\alpha}_\ell$ and productivities $\overline{A}_\ell$ are jointly set such that model-implied steady-state equilibrium populations and working-age mean earnings exactly match their empirical values *in equilibrium*. The inner loop flattens the steady-state solution and spatial calibration into a single loop over guesses for $\{\alpha_\ell, A_\ell, \rho_\ell, q_\ell\}_\ell$ jointly, conditional on housing stocks H_ℓ^S. It searches for a set of values satisfying, for each location:

1. Model populations equal data populations for all locations ℓ.
2. Model working-age (age 20–59) mean earnings equal data working-age mean earnings[14] for all locations ℓ.
3. Aggregate demand for real estate assets equals data housing supply: $H_\ell^D = H_\ell^S \ \forall \ell$.
4. Aggregate demand for rental housing equals data housing supply: $H_\ell^{\text{rent}} = H_\ell^S \ \forall \ell$.

Local housing price q_ℓ and rent ρ_ℓ levels are determined by model equilibrium and are thus untargeted. Their fit is reported in Section 3.4.4. Conditional on the model-implied prices q_ℓ, local housing supply shifters Π_ℓ are then set to,[15]

$$\Pi_\ell = \frac{H_\ell}{q_\ell^{\theta_\ell}},$$

consistent with Equation (2).[16]

13. For numerical stability, housing supply elasticities are censored below at 0.05. This affects 242 out of 2447 locations.

14. Mean earnings for households with heads aged 20–65 whose place of residence is ℓ.

15. At 3 PUMAs, the model-implied average annual disaster damages exceed the observed rents. This is not possible in the model: the housing stock would be freely disposed until rents rise to cover damages. For this reason, I do not impose the empirical housing stock target for these locations.

16. A slight tension with the data exists here. A model in which baseline prices represent a steady state is calibrated to match moments of a reality in which prices already reflect a certain degree of climate expectations. Therefore, the

Bilateral migration costs are parameterized in terms of geodesic distance,

$$F(\ell, \ell') = \left(\overline{F} + F^d d(\ell, \ell')\right) \mathbb{1}[\ell' \neq \ell].$$

and are thus determined by $\overline{F}$ and F^d, whose calibration is described with the other non-spatial parameters.

3.3 Outer Loop: Non-Spatial Parameters

5 non-spatial parameters are set internally. 5 aggregate moments are chosen, with each associated to one of the 5 parameters, with respect to which it is verified to be strongly monotonic. The five parameters are then set in the following way:

1. The full set of five parameters is guessed.
2. In an inner loop, location-level parameters are set as described in the previous subsection.
3. The simulated version of each of the five moments is computed and compared to its empirical value.
4. Each parameter is updated according to the difference between its associated moment's simulated and empirical value.
5. Repeat from Step 2 until the simulated moments equal their empirical values, within a tight tolerance.

Table 1 summarizes the calibration of non-spatial parameters. The weight on housing in consumption γ is associated with the empirical ratio of mean gross rent among renters to mean wage earnings among renters aged 20–59 in the ACS.[17] The fixed utility cost of moving $\overline{F}$ is associated with the share of households moving each decade. The per-distance utility cost of moving F^d is associated with the median distance of a county-to-county move over 20km within the contiguous United States, from the 2016–20 Census. The housing maintenance cost coefficient χ is associated with the price-to-rent ratio of the average U.S. home in 2020. The model's unit of housing is one average occupied home or 2.77 bedrooms. Prices are from Zillow and rents are the in-place gross rent reported in the ACS. The strength of the bequest motive Q is associated with mean household net worth in the 2022 SCF.

effect of the "onset shock" must be explicitly defined as the hypothetical effect of an immediate and homogeneous change in beliefs from complete climate denial to complete climate credence in the sense of correctly anticipating the full stochastic global climate process, in a world in which initial allocations are equal to 2020 allocations. Thus the model does admit a genuine estimate of the full effect of climate expectations, relative to 2020 and relative to complete climate denial, but does so in a world whose fundamental parameters generate allocations matching data in 2020 in the absence of any climate credence.

17. The ratio-of-means, not the mean-of-ratios.

Table 1—Parameter values and sources.

Parameter	Interpretation	Value	Target moment or source	Moment value	
				Model	Data
Panel A. Internally calibrated					
γ	Weight on housing services in consumption	0.02851	Ratio of mean gross rent to mean wage earnings, renters aged 20–59 (ACS 2022)	0.261	0.261
χ	Maintenance cost coefficient, share of rent	0.6538	Price-to-rent ratio of the average home, price over a decade of in-place rent (Zillow and ACS 2020)	1.486	1.484
$\overline{F}$	Fixed utility cost of moving	0.7490	Share of households moving within a decade, heads aged 20–79 (ACS 2022)	0.616	0.613
F^d/ψ	Utility cost of moving per kilometer, in utility units (F^d divided by ψ)	0.001634	Flow-weighted median distance of county-to-county moves over 20 km, in km (Census flows 2016–20)	178	178
Q	Strength of the bequest motive	2.086	Mean household net worth, thousand 2020 dollars (SCF 2022)	937	937
Panel B. Externally set					
σ	Goods–housing substitution elasticity	0.691	Albouy, Ehrlich, and Liu (2016); not separately identified from γ here.		
ψ	Migration elasticity, inverse of location-preference dispersion	4	Cross-sectional rent-on-wage slope, held fixed rather than matched.		
ξ	Housing taste-shock elasticity	2	Numerical smoothing for tractability, not a behavioral parameter		
η	Period-utility curvature (inverse of the elasticity of intertemporal substitution)	2	CRRA over the CES bundle; sensitivity has no η leg		
β	Discount factor, decadal	0.8171	0.98^{10}		
r^f	Risk-free return, decadal	0.254	SCF financial-asset returns 2001–22 (2.29% per year)		
r^m	Mortgage interest rate, decadal	0.8	6.05% per year		
κ	Equity required at purchase (loan-to-value cap 80%)	0.2	Kaplan, Mitman, and Violante (2020)		
ϕ	Realtor's fee, share of the sale price	0.07	Kaplan, Mitman, and Violante (2020)		
δ	Housing-stock depreciation, decadal	0.05	0.51% per year		
p	Idiosyncratic log-income persistence, decadal	0.737	Kaplan, Mitman, and Violante (2020), 0.97^{10}		
σ_ν	Log-income innovation standard deviation, decadal	0.56	Kaplan, Mitman, and Violante (2020), accumulated over a decade of the annual process		
ζ_a	Age shifters in log earnings		Six values, one per decade of head age, from ACS 2020 household wage earnings		
$a_{\max}$	Life-cycle periods, decades from age 20	6	Model structure		
λ_b^{init}	Entrant wealth distribution		Quartiles of SCF 2022 household net worth, downscaled to ages 20–29		
λ_z^{init}	Entrant log-income variance (normal, mean zero)	0.176	Initial persistent component of Kaplan, Mitman, and Violante (2020)		

3.4 Externally-Set Non-Spatial Parameters

The following parameters are set externally:

The discount rate is set to 0.817 decadally (0.98 annually). The interest rate r is set to 25.4% decadally (2.29% annually) to match the average return (interest and capital gains) on financial assets held by households in the period 2001-2022 in the Survey of Consumer Finances (SCF). The realtor's fee ϕ is set to 7%, the maximum loan-to-value requirement $1 - \kappa$ is set to 80%, as in Kaplan, Mitman, and Violante (2020). The interest rate on mortgages is set to 80% decadally (6.05% annually). The depreciation rate of location-level housing *stocks*[18] is set to 5% decadally (0.51% annually). The (inverse) elasticity of intertemporal substitution η is set to 2.

3.4.1 Substitutability of Housing Services with Consumption Goods

The elasticity of substitution between housing and other goods, σ, is set externally to 0.691, taken from Albouy, Ehrlich, and Liu (2016), because it is poorly separately-identified with the housing preference shifter, γ. To validate, I perform an indirect-inference exercise. I estimate the effect of rent level on rent share of household expenditure on simulated data by 2SLS, using housing supply elasticity as an instrument for housing rent, controlling for log earnings. This yields a similar estimate of $\sigma^* = 0.6545$.

3.4.2 Taste Shocks

The housing taste shock variance is set to $1/\xi = 0.5$. The variance of the location preference shock is set to the smaller $1/\psi = 1/4$ (with migration elasticity $\psi = 4$). The migration elasticity is set on the high end, to reflect a high responsiveness of migration to real wages over the decadal frequency of the model.

3.4.3 Initial Wealth Endowments

The entrant wealth distribution is piecewise linear with kinks at each quartile, matched to the mean and quartiles of the under-35 wealth distribution as reported in the 2022 Survey of Consumer Finances, deflated to 2020 dollars. Quartiles are obtained by scaling down the quartiles of the overall wealth distribution by the ratio of under-35 mean wealth to overall mean wealth in the SCF.

3.4.4 Life Cycle

A period is 10 years. Households live from age 20 to age 79: the first age group is 20-29; the last age group is 70-79. The persistence and innovation of the household income process are from Kaplan,

18. This primarily determines the threshold below which the housing stock of a location may not shrink in a single period, and is set separately from the depreciation rate or maintenance costs set by households.

Mitman, and Violante (2020), decadalized:

$$p = 0.737 = (p^{\text{KMV}})^{10} = 0.97^{10}$$
$$\sigma_\nu = 0.56 = \sqrt{(\sigma_\nu^{\text{KMV}})^2 \frac{1-(p^{\text{KMV}})^{20}}{1-(p^{\text{KMV}})^2}} = \sqrt{0.2^2 \frac{1-0.97^{20}}{1-0.97^2}}$$

The age-specific income shifters ζ_a are set to match median wage earnings by household-head age bin in the 2020 ACS.

Table 2 displays the model fit on untargeted moments.

4 Climate Damages

Direct climate damages in the model are of the following form,

$$\log \alpha_{\ell t} = \log \bar{\alpha}_\ell + \alpha_\ell^g \, \text{GSAT}_t \qquad \text{(Amenity)}$$
$$\log A_{\ell t} = \log \bar{A}_\ell + A_\ell^g \, \text{GSAT}_t \qquad \text{(Productivity)}$$
$$\delta_{\ell t}^{\text{disaster}} = \bar{\delta}_\ell + \delta_\ell^g \, \text{GSAT}_t \qquad \text{(Disaster Intensity)}$$

The climate sensitivities α_ℓ^g, A_ℓ^g, and δ_ℓ^g are allowed to vary flexibly across locations. Within locations, however, the relationship between period-t global-mean surface air temperature (GSAT_t) and local climate-sensitive fundamentals are linearized.

In this section, I describe the calibration of the climate sensitivities α_ℓ^g, A_ℓ^g, and δ_ℓ^g, and the argument for the linearization. Amenities and productivity are handled together and disasters are handled separately.

The same approach is taken for α_ℓ^g and A_ℓ^g: factorization through the temperature measures "Heating Degree Days" (HDD) and "Cooling Degree Days" (CDD). First, we show that, within each location, the relationship of HDDs with GSAT is very close to linear across an ensemble of climate scenarios, and do the same for CDDs. Second, we show that recent estimates of amenity damage functions and labour productivity damage functions are, in logs, close to being a linear combination of HDDs and CDDs. Finally, we matrix-multiply these linear relationships to obtain the univariate linear relationships assumed by the model.

For the disaster damage sensitivity δ_ℓ^g, we are more constrained by data availability. We restrict to damages to the dominant category, flooding,[19] and the only single projection which reports projected average annual losses (AAL) for combined coastal, fluvial, and pluvial flooding is Bates et al. (2021). I linearly interpolate between their AAL estimates for 2020 and 2050 under RCP4.5, in levels. I then deflate the resulting $\delta_{\ell t}^{\text{disaster}}$ by 0.655, the share of residential structures in all structures by value reported by First Street Foundation (2021).

19. Appendix C reports a more detailed breakdown.

Table 2—Untargeted moments. Prices and rents are population-weighted means over locations.

Moment	Model	Data	Source
Panel A. Homeownership rate			
All ages, not used to identify ξ	0.673	0.655	ACS 2020, owner-occupied over occupied units
Ages 30–39	0.464	0.555	ACS householders paying no cash rent
Ages 40–49	0.627	0.673	ACS householders paying no cash rent
Ages 50–59	0.724	0.745	ACS householders paying no cash rent
Ages 60–69	0.768	0.790	ACS householders paying no cash rent
Ages 70–79	0.781	0.820	ACS householders paying no cash rent
Panel B. Rent burden: annual gross rent over wage earnings, summed within the decile			
Earnings decile 1	2.179	2.976	ACS cash-renting householders aged 30–79
Earnings decile 2	1.095	0.977	ACS cash-renting householders aged 30–79
Earnings decile 3	0.707	0.643	ACS cash-renting householders aged 30–79
Earnings decile 4	0.479	0.524	ACS cash-renting householders aged 30–79
Earnings decile 5	0.423	0.438	ACS cash-renting householders aged 30–79
Earnings decile 6	0.368	0.387	ACS cash-renting householders aged 30–79
Earnings decile 7	0.293	0.339	ACS cash-renting householders aged 30–79
Earnings decile 8	0.279	0.297	ACS cash-renting householders aged 30–79
Earnings decile 9	0.269	0.258	ACS cash-renting householders aged 30–79
Earnings decile 10	0.227	0.165	ACS cash-renting householders aged 30–79
Panel C. Prices and wealth			
House price q, population-weighted, 2020, thousand 2020 dollars per home	339	303	Zillow Home Value Index 2020, household-weighted mean
Rent ρ, population-weighted, 2020, thousand 2020 dollars per home per decade	238	204	ACS 2020 in-place gross rent
Mean wealth at age 70 over age 60	1.003	1.145	SCF 2022 Bulletin Table 4, 65–74 over 55–64 mean net worth
Panel D. Housing stock where the supply shifter is set from data, millions of homes			
Big Sandy Area Development District, KY	0.070	0.054	ACS 2020 occupied bedrooms, in homes
Athens and 2 other counties, OH	0.055	0.046	ACS 2020 occupied bedrooms, in homes
Kanawha County, WV	0.067	0.058	ACS 2020 occupied bedrooms, in homes

4.1 From Global Climate to Local Conditions

The number of Heating (Cooling) Degree Days in a year is defined as the sum over the days in a year of the number of degrees $^\circ C$ by which daily temperatures are below (above) $18.3^\circ C$. For location ℓ, year y, with D_y days per year,

$$\begin{aligned}\text{HDD}_{\ell t} &= \sum_{d=1}^{D_y} \max(18.3 - \text{Temp}_{\ell dt}, 0)\\ \text{CDD}_{\ell t} &= \sum_{d=1}^{D_y} \max(\text{Temp}_{\ell dt} - 18.3, 0).\end{aligned}$$

I begin by estimating the relationship between HDD/CDD and GSAT across CMIP5 ensemble medians for each scenario and year,

$$\text{XDD}_{\ell ts} = \overline{XDD}_\ell + \text{XDD}_\ell^g \, \text{GSAT}_{ts} + e_{\ell ts}, \tag{9}$$

where XDD is one of HDD or CDD, ℓ is location, s is climate scenario.[20] Each t is 30-year bands, of which there are 9, one per decade, from 1985–2015 through 2065–2095. GSAT is the ensemble mean for each scenario-period, normalized to 0 in 2020. XDD_ℓ^g is the location-ℓ sensitivity of XDD to GSAT. Further details appear in Appendix C. The pooled R^2 for (9) is 0.991 for HDD and 0.995 for CDD.

4.2 From Local Conditions to Economic Fundamentals

To quantify the effect of HDDs and CDDs on amenities and productivity, I use the damage function estimates of Albouy et al. (2016) and Deryugina and S. M. Hsiang (2014). These damage functions, and further explanations, appear in Appendix C. With log damages on the y axis and daily temperature on the x axis, they each have a typical inverse-V shape: damages grow with distance from an ideal temperature. The slope of the right "leg" of the graph is the marginal cost of an additional degree of temperature on an already-hot day: exactly the cost of an additional CDD. Symmetrically, the slope of the left leg is exactly the (negative) cost of an additional HDD. That is, the piecewise slopes of the damage functions supply the coefficients $A^{g,\text{HDD}}$ and $A^{g,\text{CDD}}$ of the linear model,

$$\log A_{\ell t} = \log \widetilde{A}_\ell + A^{g,\text{HDD}} \text{HDD}_{\ell t} + A^{g,\text{CDD}} \text{CDD}_{\ell t} + e_{\ell t}, \tag{10}$$

and similarly for $\alpha_{\ell t}$.

20. RCP4.5 and RCP8.5.

4.3 Composing the Climate Sensitivities

Substituting equation (9) into equation (10), we obtain,

$$\log A_{\ell t} = \underbrace{\left[\log \widetilde{A}_\ell + A^{g,\text{HDD}}\overline{HDD}_\ell + A^{g,\text{CDD}}\overline{CDD}_\ell\right]}_{\log \overline{A}_\ell} + \underbrace{\left[A^{g,\text{HDD}}\text{HDD}_\ell^g + A^{g,\text{CDD}}\text{CDD}_\ell^g\right]}_{A_\ell^g} \text{GSAT}_t + \tilde{e}_{\ell t},$$

and similarly for $\alpha_{\ell t}$, so that we obtain,

$$\hat{A}_\ell^g = \hat{A}^{g,\text{HDD}}\widehat{\text{HDD}}_\ell^g + \hat{A}^{g,\text{CDD}}\widehat{\text{CDD}}_\ell^g,$$
$$\hat{\alpha}_\ell^g = \hat{\alpha}^{g,\text{HDD}}\widehat{\text{HDD}}_\ell^g + \hat{\alpha}^{g,\text{CDD}}\widehat{\text{CDD}}_\ell^g.$$

The specific papers Albouy et al. (2016) and Deryugina and S. M. Hsiang (2014) were chosen not only for their recency and compatibility with this estimating procedure, but for their compatibility with *each other*. Albouy et al. (2016) use cross-sectional variation in real wages to identify quality of life, and examine how this latent measure varies with temperature, so that their amenity measure is, by construction, net of income. Deryugina and S. M. Hsiang (2014), by contrast, use county-year weather and income data with county fixed effects. Their estimate thus uses only time-series variation to identify productivity effects of temperatures, and nets out the level effects of county-level climates.

Figure 1 displays the distribution of climate sensitivities across space. Panels (a) through (c) display the climate-sensitivities of productivity, amenities, and flood losses, respectively. Panel (d) displays baseline annual flood losses. Flood losses and flood loss sensitivities are concentrated in locations along the East coast, and to a lesser extent along the Appalachian mountains. Productivity and amenity sensitivities are largely explained by the North-South axis, because more Southern areas are hotter, and in hotter areas an additional degree of warming contributes more marginal hot days and CDDs and reduces fewer marginal cold days and HDDs. The maps of HDD and CDD responses to global temperatures across space appear in Appendix C.

Central to this estimation approach is the approximate within-location linearity of the response of $\log A_{\ell t}$ and $\log \alpha_{\ell t}$ to GSAT_t. Figure 2 plots this relationship across 18 projections, two climate scenarios, and three locations with differing climate sensitivities. The relationship is indeed highly linear within locations in the range of temperatures covered by the model: a linear fit has an R^2 of 0.9925 (productivity) and 0.9787 (amenity) across the full set of 2447 locations.

5 Climate Uncertainty

The way I construct climate damages allows me to represent aggregate climate uncertainty simply as a stochastic process for global mean temperatures GSAT_t, which are a sufficient statistic in the model for the climate-impacted location-level productivity $A_{\ell t}$, amenities $\alpha_{\ell t}$, and disaster damages $\delta_{\ell t}^{\text{disaster}}$. I construct a process anchored around two widely used climate scenarios, RCP4.5 and

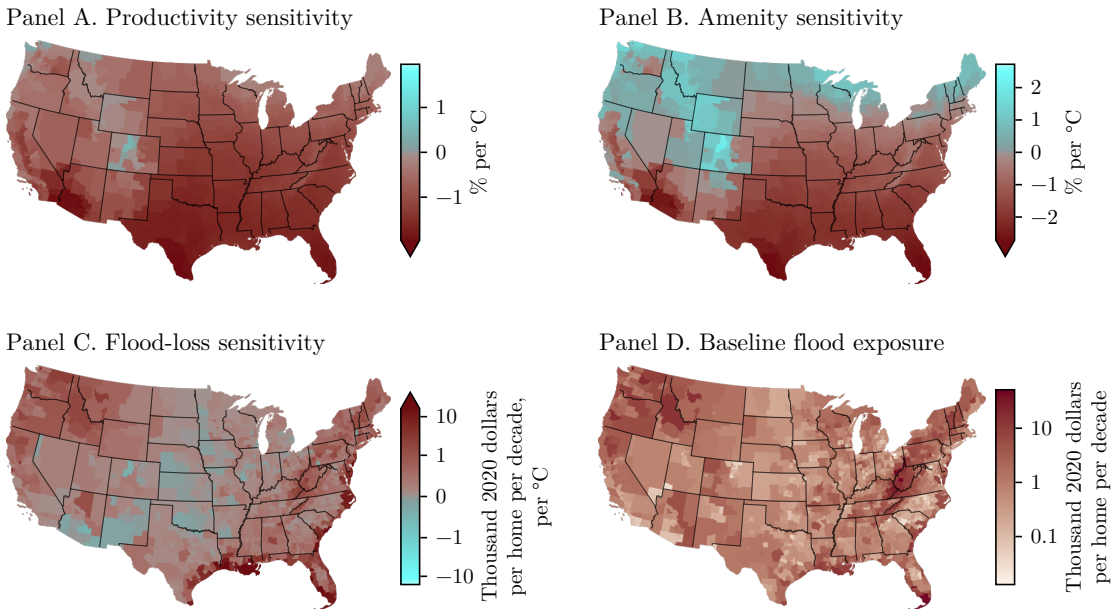


Figure 1. Climate sensitivities by location. Panel A plots the productivity sensitivity A_ℓ^g; Panel B the amenity sensitivity α_ℓ^g; Panel C the flood-loss sensitivity δ_ℓ^g; and Panel D the baseline flood loss $\bar{\delta}_\ell$. Productivity and amenity sensitivities are quantified by combining per-degree-day damage functions with degree-day projections as described in Section 4. Flood-loss levels and sensitivities are from Bates et al. (2021), as reported in Wing et al. (2022), and deflated from all-structures to residential-only by a factor of 0.655 as described in Section 4.

Locations are 2020 Public Use Microdata Areas (PUMAs), of which there are 2447 in the contiguous United States. The scales in Panels A–C are clipped at the 1st and 99th percentiles.

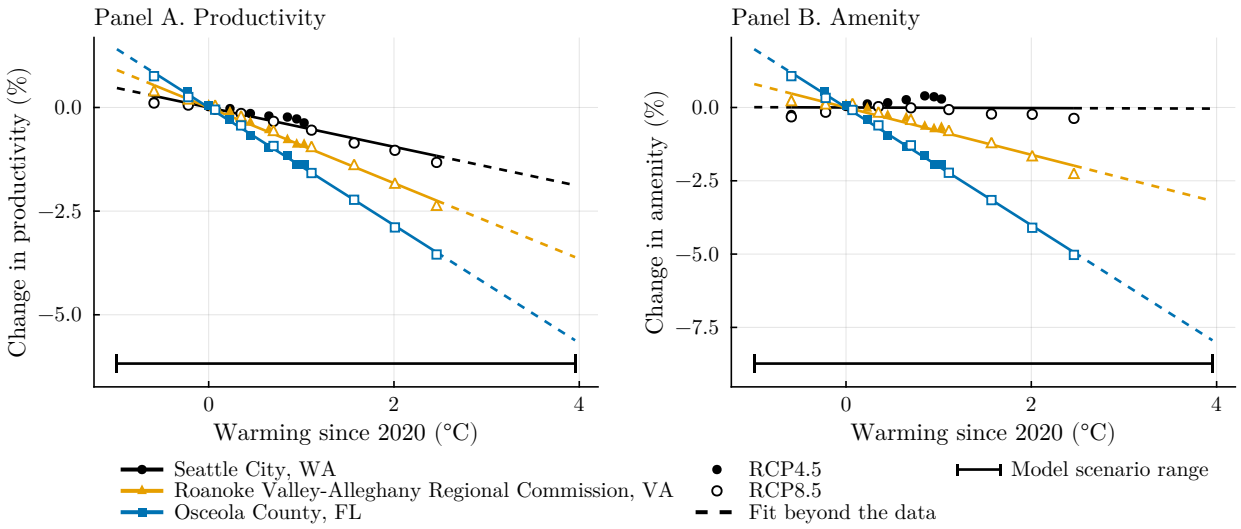


Figure 2. Relationship between damage estimates computed from linearizing degree-days with respect to GSAT and without linearizing, for three locations at the 5th, 50th, and 95th percentiles of 2020 CDDs. The slopes of the lines in Panel A are A_ℓ^g and in Panel B are α_ℓ^g. Across all 2447 locations, the R^2 of the linearizations are 0.9925 for productivity and 0.9787 for amenities. Further details are in Appendix C.

RCP6.0.

5.1 Climate Process

My goal is to specify a stochastic process for GSAT_t which is anchored by the joint evolution of the temperature projections for the benchmark climate scenarios RCP4.5 and RCP6.0 reported by Lyon et al. (2022). The scenarios' temperature projections are very tightly bunched on a 10-20 year horizon, but diverge substantially after 30 years. Although the RCP scenarios are never assigned prior probabilities by the IPCC, I use this feature of their divergence over time as an approximate quantitative reflection of how uncertainty increases with forecast horizon.

To operationalize the evolution of climate uncertainty, I construct a set of climate scenarios, indexed by $s \in \mathbb{R}$ and an AR(1) process over those scenarios. I calibrate the process so the intertercile range of year-t temperatures, given year 2020 temperatures, is equal to the difference between RCP4.5 and RCP6.0.

In particular, I construct a scenario-by-year GSAT function,

$$\text{GSAT}(t, s) = \overline{\text{GSAT}}_t + \text{GSAT}^g_t s,$$

such that GSAT is fully determined by scenario index s and time t. $\overline{\text{GSAT}}_t$ is the value at $\text{GSAT}(t, 0)$ and GSAT^g_t is a year-specific sensitivity of GSAT to the climate scenario.

I then treat s as a state variable following an AR(1) process,

$$s_{t+1} = \varrho_s s_t + \varepsilon^s, \quad \text{where} \quad \varepsilon^s \sim \mathcal{N}(0, 1 - \varrho^2) \text{ i.i.d.},$$

with the shock variance set to $1 - \varrho^2$ so that the unconditional variance of the AR(1) process is fixed at unity for any ϱ. Global mean temperature is thus given by,

$$\text{GSAT}_t = \text{GSAT}(t, s_t),$$

and all climate inputs into the model being derived from GSAT_t as described in Section 4.

I specify that the climate begins at the median scenario, $s_{2020} = 0$. This GSAT_t process then has the property that, for any value of ϱ, the conditional mean of year-t temperature, given year-2020 information, is $\overline{\text{GSAT}}_t$,

$$\mathbb{E}_{2020}[\text{GSAT}_t] = \overline{\text{GSAT}}_t.$$

I set ϱ to 0.95, so that the half-life of climate information is 135.1 years. It thus remains only to set the values of $\overline{\text{GSAT}}_t$ and GSAT^g_t. I set $\overline{\text{GSAT}}_t$ to match the midpoint of the RCP4.5 and RCP6.0 projections. I set GSAT^g_t so that the intertercile range of year-t temperature, for each t, is equal to the difference between RCP4.5 and RCP6.0.

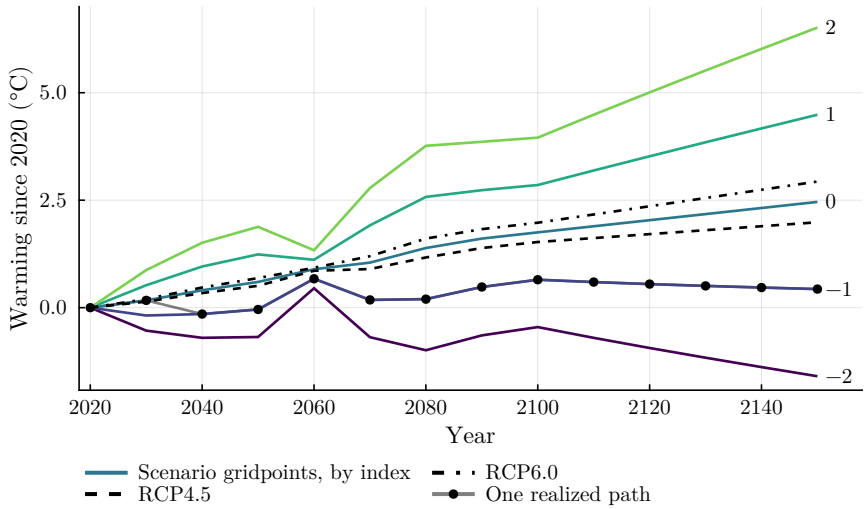


Figure 3. The gridpoints of the climate scenario indices for each year. The climate begins in 2020 at the central scenario with index 0, then evolves according to a fixed Rouwenhorst AR(1) transition matrix. The central scenario is set to the midpoint of RCP4.5 and RCP6.0 for each decade, and the variance is set so that the difference between RCP4.5 and RCP6.0 is 54.48% of the model process's *conditional variance* of GSAT each decade conditional on beginning at index 0 in 2020, as described in Section 5.

Formally, $\overline{\text{GSAT}}_t$ and GSAT_t^g are exactly identified by the equations,

$$\mathbb{E}[\text{GSAT}(t,s) \mid s_{2020} = 0] = \frac{1}{2}\left[\text{Temp}_t(\text{RCP4.5}) + \text{Temp}_t(\text{RCP6.0})\right]$$
$$\sqrt{\mathbb{E}\left[\left(\text{GSAT}(t,s) - \overline{\text{GSAT}}_t\right)^2 \mid s_{2020} = 0\right]} = \frac{\text{Temp}_t(\text{RCP6.0}) - \text{Temp}_t(\text{RCP4.5})}{\Phi^{-1}(2/3) - \Phi^{-1}(1/3)},$$

where $\text{Temp}_t(\text{RCP4.5})$ and $\text{Temp}_t(\text{RCP6.0})$ are the ensemble median year-t global mean temperatures projected by Lyon et al. (2022) under RCP4.5 and RCP6.0, respectively. Φ is the standard normal cdf.

Finally, I discretize s into five gridpoints $s \in \{-2, \ldots, 2\}$, and the AR(1) process into a time-invariant 5×5 transition matrix $\mathcal{P}$, by the Rouwenhorst method. These gridpoints are plotted in Figure 3. Further details are presented in Appendix C.5.

6 Results

This section reports the effect of climate expectations on and through housing markets in the model. The quantification of the headline effect of housing expectations on housing prices and rents across space is covered first. Stochasticity in the global climate process leads to ex ante climate uncertainty, which impacts housing prices, quantities, and rents, and has distributional consequences. I study these by comparing the model with aggregate climate uncertainty to the perfect-foresight transition.

In both the aggregate-uncertainty and perfect-foresight cases, the experiment is as follows. The economy begins at a steady-state representing 2020. All households believe that the 2020 climate will persist indefinitely. Then, all households simultaneously learn that the climate will begin to evolve according to a known, possibly stochastic, process. The initial response of housing prices is therefore exactly the estimated effect of climate expectations in the model in 2030, relative to the belief that the climate will remain forever static at the current 2020 state.[21]

Unless otherwise stated, all time-indexed results are reported along the median path, Scenario 0 as described in Section 5 (the average of RCP4.5 and RCP6.0). The perfect-foresight results presented in the main text are produced by an entirely conventional exact solver described in Appendix A.3. The model with nonzero variance in the climate process is intractable by conventional methods, so I solve it using nVFI (Sun 2026a). I perform a number of validation exercises, beginning with using nVFI to solve the perfect-foresight case and comparing with the conventional solution. Details of the solution method and a larger battery of validation tests appear in Appendix D.

As a check, nVFI is also used to re-solve the perfect-foresight model, and verified to match the conventional solver's results to a tight tolerance. The two methods differ in prices by a root-mean-square average[22] of 0.2399% and a median absolute gap across locations of 0.1188%.

6.1 Spatial Housing-Wealth Transfers

The "onset shock" — the effect of the news of the climate transition received by the initial 2020 steady state — induces a gross redistribution of housing wealth between regions of $95.58bn on impact. This is the effect of climate expectations on 2030 prices in the model, relative to the belief that the climate will remain forever at its 2020 state. The spatial reallocation grows over time to a gross redistribution of $907.8bn by 2100. The effect on housing wealth is not net-zero. The onset shock causes a decrease in the total value of housing of $−197.4bn initially and $−879.5bn by 2100.

Figure 4 maps the model-implied effect of climate expectations on housing wealth across space in 2030 and Figure 5 maps the spatial redistribution of housing wealth from 2020 to 2100.

Each of these transfers is a statement about how housing wealth moves across space. None has an interpretation as transfers between households in some locations to households in others. Following the onset shock, all climate expectations are fully priced in. Along the median path, the housing wealth reallocations in the model between 2020 and 2100 are thus fully driven by households choosing to invest in climate-protected regions over climate-vulnerable ones, for which households must pay the exact market value of those investments.

Furthermore, I do not take a stand on how climate expectations have shaped observed 2020 prices in reality. Thus, I do not take a stand on how much of the climate effect on the spatial distribution of housing wealth in 2020 is due to past price movements upon receipt of climate news, or how much is due to net-worth-neutral climate-informed past investments.

21. This is the role of *the expectation of further climate change*, not the role of climate change itself, which has already contributed to the 2020 state.

22. Root-mean-square average over locations and decades, weighted by total housing value.

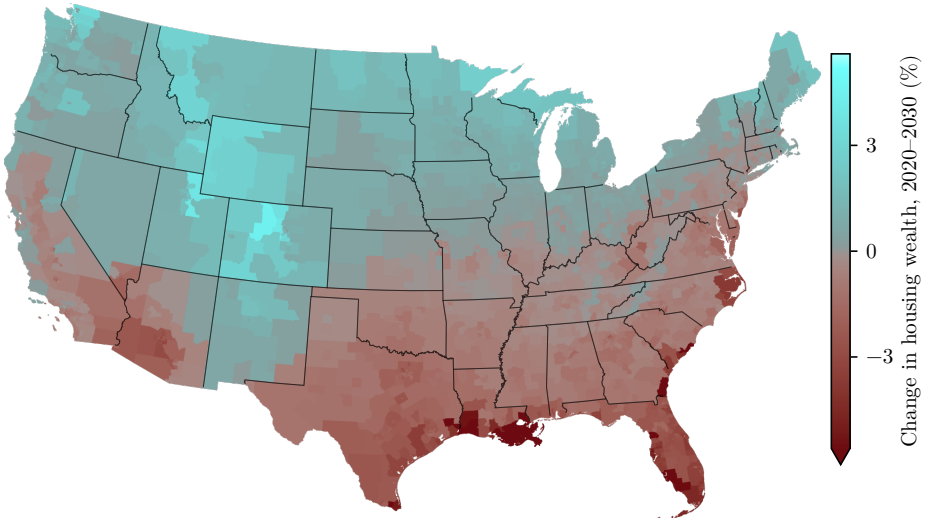


Figure 4. Effect of climate expectations on housing wealth by location in 2030.

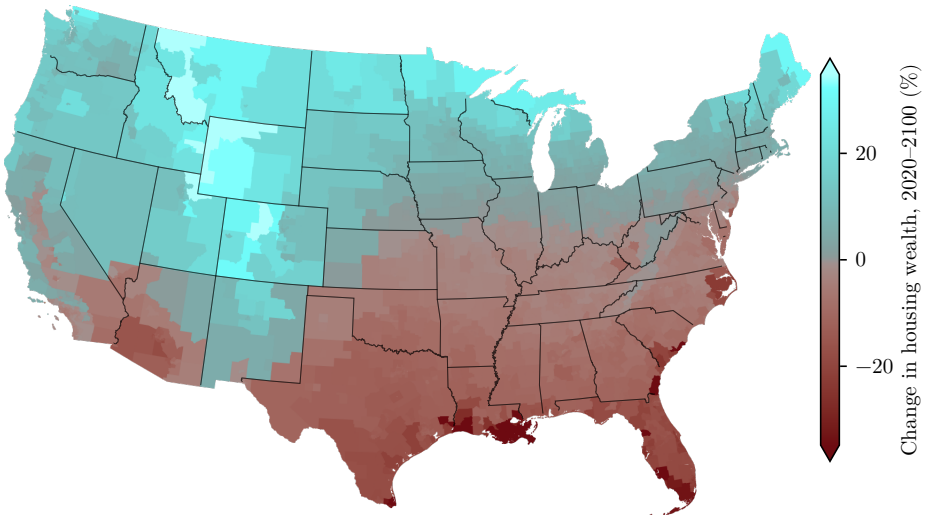


Figure 5. Change in total housing wealth in each location between 2020 and 2100 along the median path. The color scale is clipped on both ends at the 99th percentile of absolute change. The change relative to baseline can be decomposed into a net \$−879.5bn reduction of total housing wealth and a gross reallocation of \$907.8bn of housing wealth across space. Locations are 2020 Public Use Microdata Areas (PUMAs), numbering 2447. Total housing wealth is computed as $q_{\ell t} H_{\ell t}$, the product of housing price and housing stock in each location.

6.2 Distribution of Aggregate Damages, Within and Across Scenarios

How does uncertainty over future climate feed into uncertainty over future prices, damages, and housing wealth? How is that uncertainty distributed across space?

To study this, I use a numerical quadrature approach to represent the marginal distribution of these outcomes across possible climate paths. I construct a set of worse-than-median climate paths which trace out different points on the right half of the damage distributions. By symmetry, each such path has an equally-likely better-than-median counterpart obtainable by mirroring about the median scenario.

I then examine the distribution of outcomes across these drawn climate paths and across locations. Figure 6 displays the results. I show four location bins: the three terciles bins and the top decile bin.[23]

For the lower two terciles, mean prices under uncertainty are very close to the perfect-foresight prices along the median path. For the top tercile and decile, the gap widens to a peak of around half a percentage point. The spread across scenarios is much larger: the 10th to 90th percentile range across *scenarios* spans from close to zero to around twice the median price effect. The relatively small effect of the scenario spread on mean prices largely reflect two features of the model, which differ in the extent to which they reflect reality. First, the persistence of climate surprises is high and their direct damages are realized decades in the future. Price uncertainty in the present is relatively low, with a climate surprise being followed by decades of rents collected under similar physical conditions to before the surprise, their response to some extent dampened by housing investment responses. Second, household risk aversion in the model is low, with CRRA $\eta = 2$, against the $10 - 20$ typically required to generate realistic risk premia in asset pricing.

For each location bin, the median damage bill is almost identical across scenarios, by construction: the median path is the median in both, and damage bills are largely physically determined. In upper quantiles, however, the mean damage bill is lower than the median, reflecting adaptation in severe scenarios: overall damage bills are reduced in severely-impacted locations because housing investment and stocks are endogenously reduced.

Climate uncertainty induces a precautionary reallocation of housing investment from more vulnerable to less vulnerable locations, and this can be read off of the first and third columns of Figure 6. Relative to the median path, severely-impacted locations have lower prices and housing wealth before 2100, and mildly-impacted locations have higher prices and housing wealth after 2100.

6.3 In-Distribution Surprise

The global solution allows us to study the impact of climate scenario shifts when those shifts are *in-distribution* surprises. By contrast, typical perfect-foresight solutions would only model the two extremes: the fully unanticipated shift, wherein a "perfect-foresight" transition receives a shock *ex ante* believed to be impossible, and continues as another "perfect-foresight" path; and the fully

23. The top decile bin is a subset of the top tercile bin.

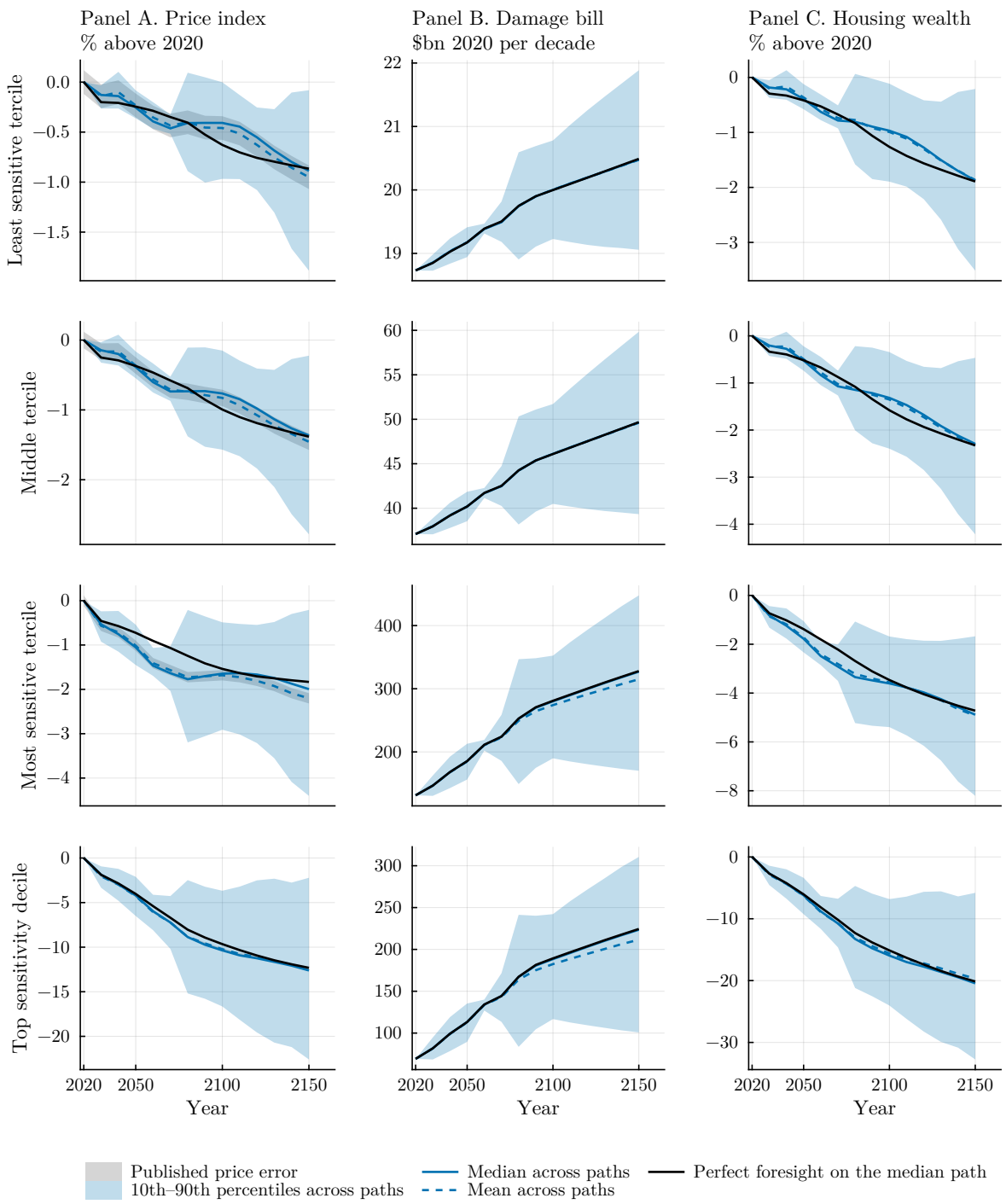


Figure 6. Outcomes by location exposure bin under the stochastic solution and the perfect-foresight median climate path. Blue shaded regions show the 10th to 90th percentiles across drawn climate paths. The median and mean across the climate paths in the global solution are shown as solid and dashed blue lines, respectively. The black line is the perfect-foresight solution on the median path. The gap between the mean and the black line is the bias of planning on the median.

Groups are terciles of the disaster damage sensitivity δ_g and its top decile. The housing price index is weighted by population. The band on the mean price line is the noise floor for the neural network solution for prices, 0.1188% of the level.

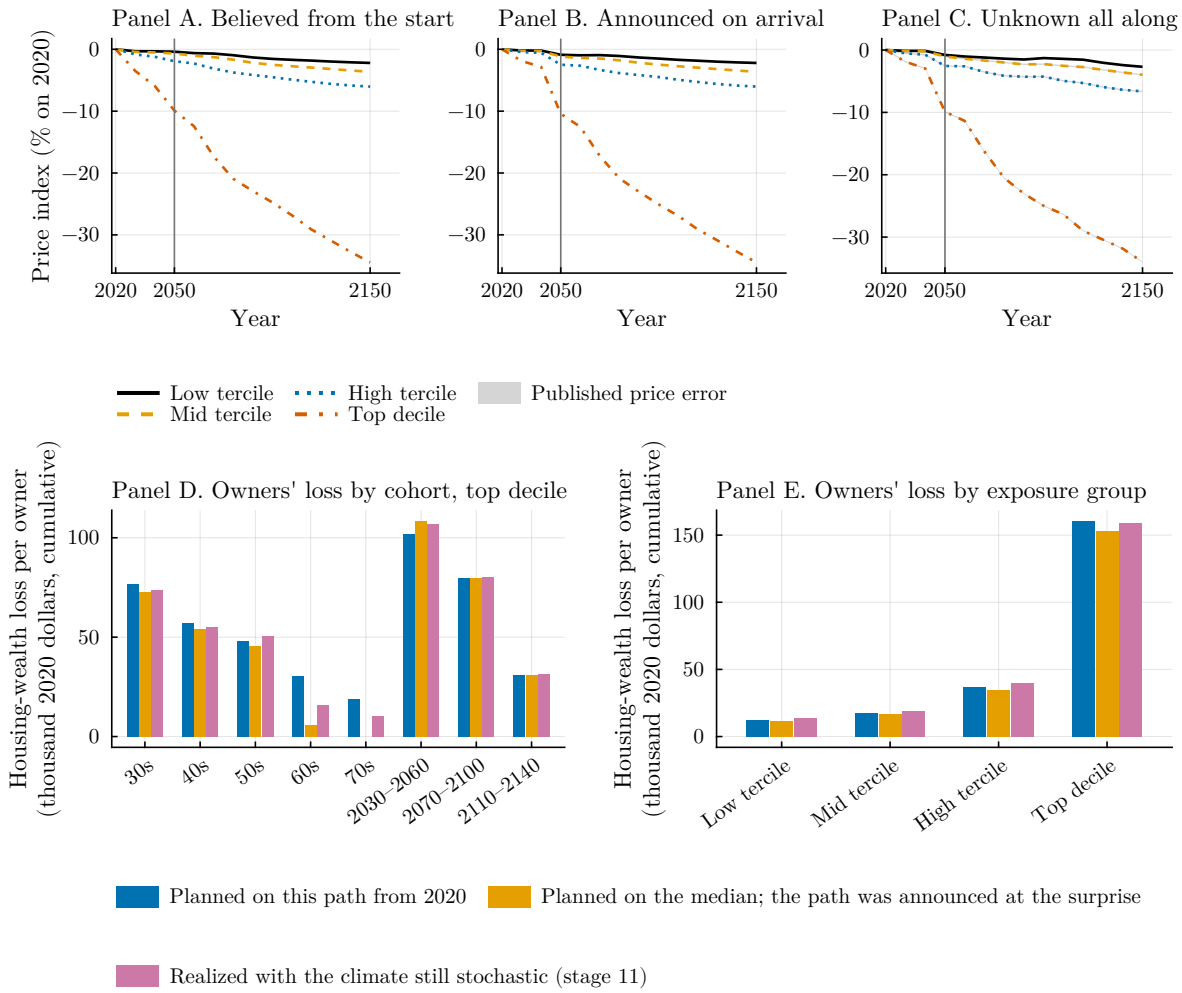


Figure 7. Prices and homeowner capital losses along one realized climate path, the median node for three periods and the bad node thereafter, under perfect foresight, a permanent shock in 2060 and perfect foresight thereafter, and the stochastic solution along the *ex post* realized path.

anticipated shift, wherein households are fully aware of the shift before hand as part of the perfect-foresight path.

Figure 7 shows the global analogue: a shift which was *ex ante* known to be *possible*, but which was not anticipated with certainty. To clarify the exercise, I analyze specifically the effect of a shift in 2060 from the median scenario (halfway between RCP4.5 and RCP6.0), to the worst of five scenarios (approximately equal to RCP8.5). For most age, cohort, and location exposure bins, the impact is between the fully-anticipated and fully-unanticipated extremes.

7 Counterfactuals

I now use the model for policy analysis. I study four policies: a "managed-retreat" policy in which land use is deregulated in flood-protected locations close to flood-exposed locations; a removal of

the effective flood-insurance subsidy provided by the National Flood-Insurance Program (NFIP); a policy designed to reduce the cost of moving for households; and a policy that improves the precision of long-run climate forecasts available to households.

All counterfactuals with the exception of the last are reported relative to the perfect-foresight case.

Managed Retreat In the "managed-retreat" policy I consider, the top 5% of locations by disaster damage are designated as "high-risk." All other locations are designated "low-risk." Low-risk locations are ordered by how close they are to the closest high-risk location. The top 5% of this ranking — the low-risk locations closest to some high-risk location — are treated with a land-use deregulation, which I model as a 5% increase in housing supply at any price level. That is, the housing supply shifter is increased by 5%.

Because this land-use deregulation would have economic benefits in the absence of a climate motive, we must isolate the effect of this policy on climate adaptation from the unconditional economic benefits it would generate in this model. For this reason, I subtract the benefit that the policy would have in the absence of climate change. Another way to conceptualize this is that I assume that a land-use deregulation has costs, unobservable to me, equal to its estimated economic benefits in the initial baseline. Then the benefit of the policy I report is equal to the benefits in the climate scenario net of those unobserved costs.

Flood-Insurance Subsidy Removal The reference perfect-foresight solution reported in Section 6 incorporates a subsidy to flood insurance of 7%, calibrated to the average effective subsidy provided by the NFIP. To study its removal, I consider a counterfactual in which this subsidy is removed, and the resulting government budget surplus is rebated lump-sum to households. To avoid confounding the estimate with the welfare effects of a wealth reallocation, I hold the wealth profile of rebate recipients equal to the wealth profile of NFIP recipients in the baseline, but keep the rebate uniform across households within wealth bins.[24]

Migration Ease To assess the climate adaptation benefit of policies which make migration easier, I consider a counterfactual in which the costs of migration are uniformly reduced by 10%. That is, both the fixed cost $\overline{F}$ and the per-kilometer cost F^d are reduced by 10%. This presents the same interpretation challenge as the land deregulation counterfactual: the change would have a positive welfare effect even in the absence of climate change. I apply the same approach of netting out the effect of the policy in the baseline steady state from the effect of the policy in the climate scenario.

Climate Forecasts To assess the climate adaptation benefit of improved climate forecasts, it is important to note that impact depends on what information the improved forecasts deliver. If the forecasts reveal that climate change will be on the severe end of the distribution, prices in the

24. Because the rebate is small relative to total household income (0.01706% of total household income in 2020), I do not re-solve the model but enter the rebate directly into the consumption-equivalent welfare calculation.

present would respond more strongly than they would to the revelation of the distribution as a whole.

That is to say, the information content of the improved climate forecast is a random variable, and therefore the effect of "improved climate forecasts" is also a random variable. I consider the simple case where the improved climate forecast collapses all uncertainty and pins down a full climate path which is then taken as perfect-foresight by households. For each path, I define the benefit of the forecast to be the difference in consumption-equivalent welfare between that perfect-foresight scenario and the global solution of the stochastic-climate model under that ex-post realized climate path.

Comparison and Analysis Table 3 reports the estimated benefits of the first three counterfactual policies in the model. The estimated net benefit of the managed-retreat policy is $\$-5.477$bn, and of the flood-insurance reform is \$19.81bn.

These numbers are not directly comparable because the policies have different sizes and cannot directly be placed on the same scale. To compare them, I estimate each policy's benefit-cost ratio (BCR). As discussed, the managed-retreat policy-benefit estimate has an interpretation as being "net of" the cost of the policy, assumed to be equal to its benefit in baseline. For the flood-insurance subsidy, I estimate the cost as the subsidy provided plus NFIP's reported expense ratio of 29.1%.[25] I can then compute the BCR of each policy as the benefit divided by the estimated cost.

Table 3 reports the (not comparable) dollarized and (comparable) BCR estimates for each policy. The estimated BCR for the flood subsidy is 0.6904, below one. The estimated BCR for the managed-retreat policy is 0.966, also slightly below one.[26]

25. Federal Emergency Management Agency (2024); Brown & Brown, Inc. (2025).

26. This is for the specific allocation of the managed retreat policy described in this paper as a simple example. A search for the optimal such allocation may yield a BCR above one, and is left for future work.

Table 3—Net effects of counterfactual policies in the model, as described in Section 7.

Counterfactual	Policy lever	Rent, 2050 (percentage points)	Spatial redistribution, 2100 (billions of 2020 dollars)
Panel A. Baseline			
Uncertainty rent premium; perfect-foresight redistribution		−0.05383	876.4
Panel B. Counterfactuals			
Managed retreat	Housing supply deregulated in locations close to highly-exposed locations	−0.1718	+8.4
Flood-insurance subsidy	Removal of 7% flood insurance subsidy in baseline	+0.02907	−5.9

Panel C. Welfare

	Net welfare, percent of consumption		Willingness to pay, billions of 2020 dollars		MVPF	
Counterfactual	Living at the onset	All generations	Net	Gross	Over the net cost	Over the subsidy alone
Managed retreat	−0.0008249	−0.001270	−5.477	+153.8	0.966	—
Flood-insurance subsidy	+0.005367	+0.004593	+19.81	−44.17	0.6904	0.8913

Effects are reported net of costs, which for managed retreat and migration-ease are set equal to their estimated benefits in the static-climate baseline steady state, and for the flood insurance subsidy is set to the overall size of the subsidy plus the 29.1% expense ratio reported by NFIP. Present values in 2030, in 2020 dollars.

References

Albouy, David, Gabriel Ehrlich, and Yingyi Liu. 2016. *Housing Demand, Cost-of-Living Inequality, and the Affordability Crisis.* Working Paper 22816. National Bureau of Economic Research.

Albouy, David, Walter Graf, Ryan Kellogg, and Hendrik Wolff. 2016. "Climate Amenities, Climate Change, and American Quality of Life." *Journal of the Association of Environmental and Resource Economists* 3 (1): 205–246. https://doi.org/10.1086/684573.

Azinovic, Marlon, Luca Gaegauf, and Simon Scheidegger. 2022. "Deep Equilibrium Nets." *International Economic Review* 63 (4): 1471–1525. https://doi.org/10.1111/iere.12575.

Bakkensen, Laura A., and Lint Barrage. 2022. "Going Underwater? Flood Risk Belief Heterogeneity and Coastal Home Price Dynamics." *The Review of Financial Studies* 35 (8): 3666–3709. https://doi.org/10.1093/rfs/hhab122.

Balboni, Clare. 2021. "In Harm's Way? Infrastructure Investments and the Persistence of Coastal Cities." Working paper.

Bansal, Ravi, Dana Kiku, and Marcelo Ochoa. 2016. *Price of Long-Run Temperature Shifts in Capital Markets.* Working Paper 22529. National Bureau of Economic Research, August. https://doi.org/10.3386/w22529.

Barnett, Michael, William Brock, and Lars Peter Hansen. 2020. "Pricing Uncertainty Induced by Climate Change." *The Review of Financial Studies* 33 (3): 1024–1066. https://doi.org/10.1093/rfs/hhz144.

Bates, Paul D., Niall Quinn, Christopher Sampson, Andrew Smith, Oliver Wing, Jeison Sosa, James Savage, et al. 2021. "Combined Modeling of US Fluvial, Pluvial, and Coastal Flood Hazard Under Current and Future Climates." *Water Resources Research* 57 (2): e2020WR028673. https://doi.org/10.1029/2020WR028673.

Baum-Snow, Nathaniel, and Lu Han. 2024. "The Microgeography of Housing Supply." *Journal of Political Economy* 132 (6): 1897–1946. https://doi.org/10.1086/728110.

Bernstein, Asaf, Matthew T. Gustafson, and Ryan Lewis. 2019. "Disaster on the Horizon: The Price Effect of Sea Level Rise." *Journal of Financial Economics* 134 (2): 253–272. https://doi.org/10.1016/j.jfineco.2019.03.013.

Besançon, Mathieu, Theodore Papamarkou, David Anthoff, Alex Arslan, Simon Byrne, Dahua Lin, and John Pearson. 2021. "Distributions.jl: Definition and Modeling of Probability Distributions in the JuliaStats Ecosystem." *Journal of Statistical Software* 98 (16): 1–30. https://doi.org/10.18637/jss.v098.i16.

Besard, Tim, Valentin Churavy, Alan Edelman, and Bjorn De Sutter. 2019. "Rapid Software Prototyping for Heterogeneous and Distributed Platforms." *Advances in Engineering Software* 132:29–46. https://doi.org/10.1016/j.advengsoft.2019.02.002.

Besard, Tim, Christophe Foket, and Bjorn De Sutter. 2019. "Effective Extensible Programming: Unleashing Julia on GPUs." *IEEE Transactions on Parallel and Distributed Systems* 30 (4): 827–841. https://doi.org/10.1109/TPDS.2018.2872064.

Bezanson, Jeff, Alan Edelman, Stefan Karpinski, and Viral B. Shah. 2017. "Julia: A Fresh Approach to Numerical Computing." *SIAM Review* 59 (1): 65–98. https://doi.org/10.1137/141000671.

Bilal, Adrien, and Esteban Rossi-Hansberg. 2023. *Anticipating Climate Change Across the United States.* Working Paper 31323. Cambridge, MA: National Bureau of Economic Research, June. https://doi.org/10.3386/w31323.

Board of Governors of the Federal Reserve System. 2023. *2022 Survey of Consumer Finances: Summary Extract Public Data [Data set].* https://www.federalreserve.gov/econres/scfindex.htm.

Bouchet-Valat, Milan, and Bogumił Kamiński. 2023. "DataFrames.jl: Flexible and Fast Tabular Data in Julia." *Journal of Statistical Software* 107 (4): 1–32. https://doi.org/10.18637/jss.v107.i04.

Brown & Brown, Inc. 2025. *Form 10-K for the Fiscal Year Ended December 31, 2024.* U.S. Securities and Exchange Commission, Note 16, "Insurance Company Subsidiary Operations". https://www.sec.gov/Archives/edgar/data/79282/000095017025019039/bro-20241231.htm.

Cai, Yongyang, and Thomas S. Lontzek. 2019. "The Social Cost of Carbon with Economic and Climate Risks." *Journal of Political Economy* 127 (6): 2684–2734. https://doi.org/10.1086/701890.

Carleton, Tamma, Amir Jina, Michael Delgado, Michael Greenstone, Trevor Houser, Solomon Hsiang, Andrew Hultgren, et al. 2022. "Valuing the Global Mortality Consequences of Climate Change Accounting for Adaptation Costs and Benefits." *The Quarterly Journal of Economics* 137 (4): 2037–2105. https://doi.org/10.1093/qje/qjac020.

Center for Emergency Management and Homeland Security. *Spatial Hazard Events and Losses Database for the United States [Online Database].* Arizona State University. https://sheldus.org.

Crews, Levi. 2023. "A Dynamic Spatial Knowledge Economy." Working paper.

Cruz, José-Luis, and Esteban Rossi-Hansberg. 2024. "The Economic Geography of Global Warming." *The Review of Economic Studies* 91 (2): 899–939. https://doi.org/10.1093/restud/rdad042.

Danisch, Simon, and Julius Krumbiegel. 2021. "Makie.jl: Flexible High-Performance Data Visualization for Julia." *Journal of Open Source Software* 6 (65): 3349. https://doi.org/10.21105/joss.03349.

den Haan, Wouter J., and Marina Marcet. 1990. "Solving the Stochastic Growth Model by Parameterizing Expectations." *Journal of Business and Economic Statistics* 8 (1): 31–34. https://doi.org/10.1080/07350015.1990.10509770.

Deryugina, Tatyana, and Solomon Hsiang. 2017. *The Marginal Product of Climate.* Working Paper 24072. Cambridge, MA: National Bureau of Economic Research, November. https://doi.org/10.3386/w24072.

Deryugina, Tatyana, and Solomon M. Hsiang. 2014. *Does the Environment Still Matter? Daily Temperature and Income in the United States.* Working Paper 20750. Cambridge, MA: National Bureau of Economic Research, December. https://doi.org/10.3386/w20750.

Desmet, Klaus, Robert E. Kopp, Scott A. Kulp, Dávid Krisztián Nagy, Michael Oppenheimer, Esteban Rossi-Hansberg, and Benjamin H. Strauss. 2021. "Evaluating the Economic Cost of Coastal Flooding." *American Economic Journal: Macroeconomics* 13 (2): 444–486. https://doi.org/10.1257/mac.20180366.

Favilukis, Jack, Sydney C. Ludvigson, and Stijn Van Nieuwerburgh. 2017. "The Macroeconomic Effects of Housing Wealth, Housing Finance, and Limited Risk Sharing in General Equilibrium." *Journal of Political Economy* 125 (1): 140–223. https://doi.org/10.1086/689606.

Federal Emergency Management Agency. 2024. *Financial Assistance/Subsidy Arrangement: Write Your Own Arrangement for Fiscal Year 2025.* Federal Register 89 FR 22420, Article IV.B, April. https://www.govinfo.gov/content/pkg/FR-2024-04-01/pdf/2024-06805.pdf.

Fernández-Villaverde, Jesús, Galo Nuño, George Sorg-Langhans, and Maximilian Vogler. 2020. "Solving High-Dimensional Dynamic Programming Problems using Deep Learning." Working paper.

First Street Foundation. 2021. *The Cost of Climate: America's Growing Flood Risk.* Technical report. Brooklyn, NY: First Street Foundation, February. https://firststreet.org/research-library/the-cost-of-climate.

Fleischmann, Martin, Joris Van den Bossche, Kelsey Jordahl, Matt Richards, James McBride, Jeff Wasserman, Brendan Ward, and Levi John Wolf. 2026. "GeoPandas: Fundamental Data Structures for Vector Spatial Data in Python." *Computers, Environment and Urban Systems* 130:102495. https://doi.org/10.1016/j.compenvurbsys.2026.102495.

Fried, Stephie. 2022. "Seawalls and Stilts: A Quantitative Macro Study of Climate Adaptation." *The Review of Economic Studies* 89 (6): 3303–3344. https://doi.org/10.1093/restud/rdab099.

Gassert, Francis, Enrique Cornejo, and Emily Nilson. 2021. "Making Climate Data Accessible: Methods for Producing NEX-GDDP and LOCA Downscaled Climate Indicators." Working paper. https://doi.org/10.46830/writn.19.00117.

GDAL/OGR contributors. *GDAL/OGR Geospatial Data Abstraction Software Library.* Open Source Geospatial Foundation. https://doi.org/10.5281/zenodo.5884351. https://gdal.org.

GEOS contributors. 2025. *GEOS Computational Geometry Library.* Open Source Geospatial Foundation. https://doi.org/10.5281/zenodo.11396894. https://libgeos.org.

Giannone, Elisa, Nuno Paixão, Xinle Pang, and Qi Li. 2023. "A Quantitative Spatial Equilibrium Model with Wealth." Working paper.

Gillies, Sean, and Rasterio contributors. 2013. *Rasterio: Geospatial Raster I/O for Python Programmers.* Mapbox. https://github.com/rasterio/rasterio.

Gillies, Sean, Casper van der Wel, Joris Van den Bossche, Mike W. Taves, Joshua Arnott, Brendan C. Ward, and Shapely contributors. *Shapely.* Zenodo. https://doi.org/10.5281/zenodo.5597138.

Golosov, Mikhail, John Hassler, Per Krusell, and Aleh Tsyvinski. 2014. "Optimal Taxes on Fossil Fuel in General Equilibrium." *Econometrica* 82 (1): 41–88. https://doi.org/10.3982/ECTA10217.

Greaney, Brian. 2023. "Homeownership and the Distributional Effects of Uneven Regional Growth." Working paper. https://doi.org/10.2139/ssrn.4406024.

Gu, Zhouzhou, Mathieu Laurière, Sebastian Merkel, and Jonathan Payne. 2024. "Global Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models." Working paper. https://doi.org/10.2139/ssrn.4871228.

Han, Jiequn, Yucheng Yang, and Weinan E. 2021. "DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks." Working paper. https://doi.org/10.2139/ssrn.3990409.

Harris, Charles R., K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, et al. 2020. "Array Programming with NumPy." *Nature* 585 (7825): 357–362. https://doi.org/10.1038/s41586-020-2649-2.

Hsiang, Solomon, Robert Kopp, Amir Jina, James Rising, Michael Delgado, Shashank Mohan, D. J. Rasmussen, et al. 2017. "Estimating economic damage from climate change in the United States." *Science* 356 (6345): 1362–1369. https://doi.org/10.1126/science.aal4369.

Hsiao, Allan. 2023. "Sea Level Rise and Urban Adaptation in Jakarta." Working paper, Stanford University. https://allanhsiao.com/files/Hsiao_jakarta.pdf.

Innes, Michael. 2018a. "Don't Unroll Adjoint: Differentiating SSA-Form Programs." *CoRR* abs/1810.07951. https://arxiv.org/abs/1810.07951.

Innes, Michael. 2018b. "Flux: Elegant Machine Learning with Julia." *Journal of Open Source Software* 3 (25): 602. https://doi.org/10.21105/joss.00602.

Innes, Michael, Elliot Saba, Keno Fischer, Dhairya Gandhi, Marco Concetto Rudilosso, Neethu Mariya Joy, Tejan Karmali, Avik Pal, and Viral Shah. 2018. "Fashionable Modelling with Flux." *CoRR* abs/1811.01457. https://arxiv.org/abs/1811.01457.

Intergovernmental Panel on Climate Change. 2013. *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change.* Edited by Thomas F. Stocker, Dahe Qin, Gian-Kasper Plattner, Melinda Tignor, Simon K. Allen, Judith Boschung, Alexander Nauels, Yu Xia, Vincent Bex, and Pauline M. Midgley. Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press. https://doi.org/10.1017/CBO9781107415324.

IPUMS USA. *IPUMS USA [Data set].* IPUMS, University of Minnesota. https://usa.ipums.org/usa/.

Kaplan, Greg, Kurt Mitman, and Giovanni L. Violante. 2020. "The Housing Boom and Bust: Model Meets Evidence." *Journal of Political Economy* 128 (9): 3285–3345. https://doi.org/10.1086/708816.

Kleinman, Benny, Ernest Liu, and Stephen J Redding. 2023. "Dynamic Spatial General Equilibrium." Working paper.

Komissarova, Kristina. 2022. "Location Choices over the Life Cycle: The Role of Relocation for Retirement." Working paper.

Krusell, Per, and Anthony A. Smith Jr. 1998. "Income and Wealth Heterogeneity in the Macroeconomy." *Journal of Political Economy* 106 (5): 867–896. https://doi.org/10.1086/250034.

Krusell, Per, and Anthony A. Smith Jr. 2022. "Climate Change Around the World." Working paper.

Lemoine, Derek, and Christian Traeger. 2014. "Watch Your Step: Optimal Policy in a Tipping Climate." *American Economic Journal: Economic Policy* 6 (1): 137–166. https://doi.org/10.1257/pol.6.1.137.

Lyon, Christopher, Erin E. Saupe, Christopher J. Smith, Daniel J. Hill, Andrew P. Beckerman, Lindsay C. Stringer, Robert Marchant, et al. 2022. "Climate change research and action must look beyond 2100." *Global Change Biology* 28 (2): 349–361. https://doi.org/10.1111/gcb.15871.

Maliar, Lilia, Serguei Maliar, and Pablo Winant. 2021. "Deep learning for solving dynamic economic models." *Journal of Monetary Economics* 122:76–101. https://doi.org/10.1016/j.jmoneco.2021.07.004.

Manson, Steven, Jonathan Schroeder, David Van Riper, and Steven Ruggles. 2021. *IPUMS National Historical Geographic Information System: Version 16.0 [Data set].* IPUMS, University of Minnesota. https://doi.org/10.18128/D050.V16.0. https://www.nhgis.org.

McKinney, Wes. 2010. "Data Structures for Statistical Computing in Python." In *Proceedings of the 9th Python in Science Conference,* edited by Stéfan van der Walt and Jarrod Millman, 56–61. https://doi.org/10.25080/Majora-92bf1922-00a.

Murfin, Justin, and Matthew Spiegel. 2020. "Is the Risk of Sea Level Rise Capitalized in Residential Real Estate?" *The Review of Financial Studies* 33 (3): 1217–1255. https://doi.org/10.1093/rfs/hhz134.

NASA Center for Climate Simulation. *NASA Earth Exchange Global Daily Downscaled Projections (NEX-GDDP), CMIP5 [Data set].* National Aeronautics and Space Administration. https://www.nccs.nasa.gov/services/data-collections/land-based-products/nex-gddp.

National Bureau of Economic Research. 2010. *County Distance Database: 2010 Census SF1 County Distances (Miles) [Data set].* https://www.nber.org/research/data/county-distance-database.

Pang, Xinle, and Pin Sun. 2023. "Moving into Risky Floodplains." Working paper.

PROJ contributors. *PROJ Coordinate Transformation Software Library.* Open Source Geospatial Foundation. https://doi.org/10.5281/zenodo.5884394. https://proj.org.

Rode, Ashwin, Tamma Carleton, Michael Delgado, Michael Greenstone, Trevor Houser, Solomon Hsiang, Andrew Hultgren, et al. 2021. "Estimating a social cost of carbon for global energy consumption." *Nature* 598 (7880): 308–314. https://doi.org/10.1038/s41586-021-03883-8.

Rouwenhorst, K. Geert. 1995. "Asset Pricing Implications of Equilibrium Business Cycle Models." Chap. 10 in *Frontiers of Business Cycle Research,* edited by Thomas F. Cooley, 294–330. Princeton, NJ: Princeton University Press. https://doi.org/10.1515/9780691218052-014.

Saiz, Albert. 2010. "The Geographic Determinants of Housing Supply." *The Quarterly Journal of Economics* 125 (3): 1253–1296. https://doi.org/10.1162/qjec.2010.125.3.1253.

Sinai, Todd, and Nicholas S. Souleles. 2005. "Owner-Occupied Housing as a Hedge Against Rent Risk." *The Quarterly Journal of Economics* 120 (2): 763–789. https://doi.org/10.1093/qje/120.2.763.

Singhvi, Anshul, Rafael Schouten Arnott, and GeometryOps contributors. 2023. *GeometryOps.jl.* https://github.com/JuliaGeo/GeometryOps.jl.

Sun, Jeffrey E. 2026a. "Continuation Value is All You Need." Working paper.

Sun, Jeffrey E. 2026b. "Household Stages." Working paper.

The pandas development team. 2020. *pandas-dev/pandas: Pandas.* Zenodo. https://doi.org/10.5281/zenodo.3509134.

Thrasher, Bridget, Edwin P. Maurer, Chris McKellar, and Philip B. Duffy. 2012. "Technical Note: Bias correcting climate model simulated daily temperature extremes with quantile mapping." *Hydrology and Earth System Sciences* 16 (9): 3309–3314. https://doi.org/10.5194/hess-16-3309-2012.

U.S. Census Bureau. 2020a. *2020 Census Tract to 2020 PUMA Relationship File [Data set].* Accessed August 1, 2026. https://www2.census.gov/geo/docs/maps-data/data/rel2020/2020_Census_Tract_to_2020_PUMA.txt.

U.S. Census Bureau. 2020b. *County-to-County Migration Flows: 2016–2020 ACS [Data set].* Accessed January 19, 2026. https://www.census.gov/data/tables/2020/demo/geographic-mobility/county-to-county-migration-2016-2020.html.

U.S. Census Bureau. 2021. *Housing Units, Table B25001, American Community Survey 5-Year Estimates Detailed Tables [Data set].* Accessed August 1, 2026. https://data.census.gov/table/ACSDT5Y2021.B25001.

U.S. Census Bureau. 2022. *2022 American Community Survey: 1-Year Estimates — Public Use Microdata Sample [Data set].* Accessed July 18, 2026. https://www.census.gov/programs-surveys/acs/microdata.html.

Wing, Oliver E. J., William Lehman, Paul D. Bates, Christopher C. Sampson, Niall Quinn, Andrew M. Smith, Jeffrey C. Neal, Jeremy R. Porter, and Carolyn Kousky. 2022. "Inequitable patterns of US flood risk in the Anthropocene." *Nature Climate Change* 12 (2): 156–162. https://doi.org/10.1038/s41558-021-01265-6.

Yao, Rui, and Harold H. Zhang. 2005. "Optimal Consumption and Portfolio Choices with Risky Housing and Borrowing Constraints." *Review of Financial Studies* 18 (1): 197–239. https://doi.org/10.1093/rfs/hhh007.

Zillow. *Zillow Home Value Index (ZHVI), All Homes, Smoothed, Seasonally Adjusted, County [Data set].* Zillow Research. https://www.zillow.com/research/data/.

A Full Household's Problem

This appendix provides the full recursive formulation of the household's within-period optimization problem, decomposed into the stages described in Section 2.1.9. For each stage, I define the associated value function (solved backward from V_a^{end} to V_a^{start}) and the forward transition of the household state distribution. The age subscript a and aggregate state Γ_t are suppressed where clear from context.

At any point within the period, the household's individual state is $(b, z, h^{\text{own}}, \ell)$, where b denotes bond holdings (negative if mortgaged), z is the idiosyncratic labor endowment, $h^{\text{own}} \in \mathcal{H} = \{0, h_1, \ldots, h_{N_H}\}$ is owned housing (0 for renters), and $\ell \in \mathcal{L}$ is the location of residence. I define intermediate value functions $V^{\text{postprice}}$, V^{postsell}, V^{postmove}, $V^{\text{postincome}}$ in addition to those introduced in the main text (V^{start}, V^{postbuy}, $V^{\text{postconsume}}$, V^{end}).

The full recursive system linking V_a^{start} to V_a^{end} within a period is:

$$V_a^{\text{start}}(b, z, h, \ell) = V_a^{\text{postprice}}(b, z, h, \ell)$$

$$V_a^{\text{postprice}}(b, z, h, \ell) = \max_{\mathbb{1}^{\text{sell}} \in \{0,1\}} V_a^{\text{postsell}}\big(b + \mathbb{1}^{\text{sell}}(1-\phi)\, q_{\ell t}\, h,\ z,\ (1-\mathbb{1}^{\text{sell}})h,\ \ell\big) \tag{11a}$$
$$\text{s.t.} \quad \mathbb{1}^{\text{sell}} = 1 \ \text{ if } \ b < -(1-\kappa)\, q_{\ell t}\, h$$

$$V_a^{\text{postsell}}(b, z, 0, \ell) = \psi^{-1} \log \sum_{\ell' \in \mathcal{L}} \exp\Big(\psi\big[V_a^{\text{postmove}}(b, z, 0, \ell') - F(\ell, \ell')\big]\Big) \tag{11b}$$

$$V_a^{\text{postmove}}(b, z, 0, \ell) = \xi^{-1} \log \sum_{h' \in \mathcal{H}} \exp\Big(\xi\, V_a^{\text{postbuy}}\big(b - q_{\ell t}\, h',\ z,\ h',\ \ell\big)\Big) \tag{11c}$$

$$V_a^{\text{postbuy}}(\tilde{b}, z, h, \ell) = V_a^{\text{postincome}}\big(\tilde{b} + \text{Net Income}(\tilde{b}, z, h, \ell),\ z,\ h,\ \ell\big) \tag{11d}$$

$$V_a^{\text{postincome}}(w, z, h, \ell) = \max_{\substack{g \geq 0 \\ h^{\text{rent}} \geq 0}} u(g, h^{\text{rent}}, \alpha_{\ell t}) + V_a^{\text{postconsume}}(b_{t+1},\ z,\ h,\ \ell) \tag{11e}$$
$$\text{s.t.} \quad b_{t+1} = w - g - \rho_{\ell t}\, h^{\text{rent}}, \quad b_{t+1} \geq -(1-\kappa)\, q_{\ell t}\, h$$

$$V_a^{\text{postconsume}}(b, z, h, \ell) = \beta \sum_{z'} \pi_a(z' \mid z)\, V_a^{\text{end}}(b,\ z',\ h,\ \ell). \tag{11f}$$

Equations (11b)–(11c) apply to renters ($h = 0$). Homeowners ($h > 0$) cannot migrate or purchase additional housing, so $V_a^{\text{postsell}} = V_a^{\text{postmove}} = V_a^{\text{postbuy}}$ for $h > 0$, collapsing Stages 4 and 5 to an identity. The boundary conditions completing the recursion across ages are:

$$V_a^{\text{end}}(b, z, h, \ell;\, \Gamma) = \mathbb{E}_{\Gamma'|\Gamma}\big[V_{a+1}^{\text{start}}(b, z, h, \ell;\, \Gamma')\big] \qquad \text{for } a < a_{\max},$$
$$V_{a_{\max}}^{\text{end}}(b, z, h, \ell;\, \Gamma) = Q\, u(\tilde{b}^{\text{liq}}, \rho_{\ell t}) \qquad \text{(bequest utility)},$$

where the bequest value is as defined in Section 2 and the expectation is taken over the aggregate transition $\Omega(\Gamma' \mid \Gamma)$.

I now elaborate on each stage, providing the economic interpretation, choice probabilities (where applicable), and the associated transition of the household state distribution λ.

Stage 2: House Price Revaluation

Current-period housing prices $q_{\ell t}$ and rents $\rho_{\ell t}$ are observed. No household decision is made. The household's bond position b and housing stock h are unchanged; only the applicable prices in all subsequent constraints and continuation values are updated. In the bond representation, this stage is an identity:

$$V_a^{\text{start}}(b, z, h, \ell) = V_a^{\text{postprice}}(b, z, h, \ell).$$

The household distribution is unchanged: $\lambda^{\text{postprice}} = \lambda^{\text{start}}$.

Stage 3: Sell Decision

Each homeowner ($h > 0$) chooses whether to sell all owned housing ($\mathbb{1}^{\text{sell}} = 1$) or retain it ($\mathbb{1}^{\text{sell}} = 0$). Selling yields $(1 - \phi)\, q_{\ell t}\, h$ in bond proceeds, leaving the household with no housing. A household whose current bond-to-housing position violates the loan-to-value constraint at updated prices,

$$b < -(1 - \kappa)\, q_{\ell t}\, h,$$

is forced to sell. Renters ($h = 0$) face no decision. The value function is equation (11a):

$$V_a^{\text{postprice}}(b, z, h, \ell) = \max_{\mathbb{1}^{\text{sell}} \in \{0,1\}} V_a^{\text{postsell}}\big(b + \mathbb{1}^{\text{sell}}(1 - \phi)\, q_{\ell t}\, h,\ z,\ (1 - \mathbb{1}^{\text{sell}})h,\ \ell\big).$$

The optimal sell policy $\mathbb{1}^{\text{sell}*}(b, z, h, \ell) \in \{0, 1\}$ is deterministic. In the forward distribution transition, each homeowner either retains its state (if $\mathbb{1}^{\text{sell}*} = 0$) or transitions to a renter state with updated bonds (if $\mathbb{1}^{\text{sell}*} = 1$):

$$(b, z, h, \ell) \xrightarrow{\mathbb{1}^{\text{sell}*}=1} \big(b + (1 - \phi)\, q_{\ell t}\, h,\ z,\ 0,\ \ell\big).$$

Stage 4: Migration

Renters draw i.i.d. location preference shocks $\varepsilon_{\ell'} \sim \text{Gumbel}(0, \psi^{-1})$ for each candidate destination $\ell' \in \mathcal{L}$ and choose their new location. Homeowners ($h > 0$), who are tied to their location by owned housing, pass through this stage without change: $V_a^{\text{postsell}}(b, z, h, \ell) = V_a^{\text{postmove}}(b, z, h, \ell)$ for $h > 0$.

Migrating from ℓ to ℓ' incurs a utility cost $F(\ell, \ell') \geq 0$ with $F(\ell, \ell) = 0$. The logit inclusive value for renters is equation (11b):

$$V_a^{\text{postsell}}(b, z, 0, \ell) = \psi^{-1} \log \sum_{\ell' \in \mathcal{L}} \exp\Big(\psi\big[V_a^{\text{postmove}}(b, z, 0, \ell') - F(\ell, \ell')\big]\Big).$$

The migration probability from origin ℓ to destination ℓ' is:

$$P^{\text{move}}(\ell' \mid b, z, \ell) = \frac{\exp\Big(\psi\big[V_a^{\text{postmove}}(b, z, 0, \ell') - F(\ell, \ell')\big]\Big)}{\sum_{\ell'' \in \mathcal{L}} \exp\Big(\psi\big[V_a^{\text{postmove}}(b, z, 0, \ell'') - F(\ell, \ell'')\big]\Big)}.$$

The distribution of renters is redistributed across locations:

$$\lambda^{\text{postmove}}(b, z, 0, \ell') = \sum_{\ell \in \mathcal{L}} P^{\text{move}}(\ell' \mid b, z, \ell)\ \lambda^{\text{postsell}}(b, z, 0, \ell).$$

Homeowners pass through: $\lambda^{\text{postmove}}(b, z, h, \ell) = \lambda^{\text{postsell}}(b, z, h, \ell)$ for $h > 0$.

Stage 5: Housing Purchase

Renters ($h = 0$) draw i.i.d. real estate taste shocks $\varepsilon^h_{h'} \sim \text{Gumbel}(0, \xi^{-1})$ for each housing level $h' \in \mathcal{H}$ and choose their housing. Choosing $h' = 0$ means remaining a renter; choosing $h' > 0$ means purchasing h' units at price $q_{\ell t}$, which decreases bonds by $q_{\ell t}\, h'$. Homeowners pass through unchanged: $V_a^{\text{postmove}}(b, z, h, \ell) = V_a^{\text{postbuy}}(b, z, h, \ell)$ for $h > 0$.

The logit inclusive value for renters is equation (11c):

$$V_a^{\text{postmove}}(b, z, 0, \ell) = \xi^{-1} \log \sum_{h' \in \mathcal{H}} \exp\Big(\xi\, V_a^{\text{postbuy}}\big(b - q_{\ell t}\, h',\ z,\ h',\ \ell\big)\Big).$$

Housing levels for which the continuation value V_a^{postbuy} is $-\infty$ (because downstream borrowing constraints cannot be satisfied) are effectively excluded from the choice set.

The purchase probability is:

$$P^{\text{buy}}(h' \mid b, z, \ell) = \frac{\exp\Big(\xi\, V_a^{\text{postbuy}}\big(b - q_{\ell t}\, h',\ z,\ h',\ \ell\big)\Big)}{\sum_{h'' \in \mathcal{H}} \exp\Big(\xi\, V_a^{\text{postbuy}}\big(b - q_{\ell t}\, h'',\ z,\ h'',\ \ell\big)\Big)}.$$

In the forward transition, renter mass at $(b, z, 0, \ell)$ is allocated across housing levels h' with probability $P^{\text{buy}}(h' \mid b, z, \ell)$, and each buyer's bond position shifts to $b - q_{\ell t}\, h'$. Homeowner mass passes through unchanged.

Stage 6: Income

All households receive income. No decision is made. Bond holdings are augmented by net income:

$$\tilde{b}^{\text{postinc}} = \tilde{b} + \text{Net Income}(\tilde{b}, z, h, \ell),$$

where $\tilde{b}$ denotes the bond position after all housing transactions (Stages 3–5) and

$$\text{Net Income}(\tilde{b}, z, h, \ell) = \underbrace{A_{\ell t}\, z}_{\text{labor}} + \underbrace{\left(r^f \mathbb{1}[\tilde{b} \geq 0] + r^m \mathbb{1}[\tilde{b} < 0]\right) \tilde{b}}_{\text{bond interest}} + \underbrace{\left((1-\chi)\, \rho_{\ell t} - \delta_{\ell t}^{\text{insurance}}\right) h}_{\text{net rental income}}.$$

The value function is equation (11d):

$$V_a^{\text{postbuy}}(\tilde{b}, z, h, \ell) = V_a^{\text{postincome}}\big(\tilde{b} + \text{Net Income}(\tilde{b}, z, h, \ell),\ z,\ h,\ \ell\big).$$

The forward distribution transition deterministically shifts each household's bond position by its net income.

Stage 7: Consumption and Savings

The household chooses consumption of non-housing goods g and housing services h^{rent} to maximize flow utility plus the continuation value, subject to a budget constraint and the borrowing constraint. Writing $w = \tilde{b} + \text{Net Income}$ for liquid resources, the problem is equation (11e):

$$\begin{aligned} V_a^{\text{postincome}}(w, z, h, \ell) = \max_{\substack{g \geq 0 \\ h^{\text{rent}} \geq 0}} \ & u(g, h^{\text{rent}}, \alpha_{\ell t}) + V_a^{\text{postconsume}}(b_{t+1},\ z,\ h,\ \ell) \\ \text{s.t.} \quad & b_{t+1} = w - g - \rho_{\ell t}\, h^{\text{rent}} \\ & b_{t+1} \geq -(1-\kappa)\, q_{\ell t}\, h. \end{aligned}$$

Under the CES specification for u, the optimal allocation of expenditure $c \equiv g + \rho_{\ell t}\, h^{\text{rent}}$ across goods and housing services is:

$$g^* = (1-\gamma)^\sigma \frac{c}{P_{\ell t}^{1-\sigma}}, \qquad\qquad h^{\text{rent}*} = \left(\frac{\gamma}{\rho_{\ell t}}\right)^\sigma \frac{c}{P_{\ell t}^{1-\sigma}},$$

where $P_{\ell t} = \left((1-\gamma)^\sigma + \gamma^\sigma\, \rho_{\ell t}^{1-\sigma}\right)^{1/(1-\sigma)}$ is the CES price index. The problem then reduces to a one-dimensional choice over total expenditure c (or equivalently, savings $b_{t+1} = w - c$). The optimal savings policy $b_{t+1}^*(w, z, h, \ell)$ is deterministic. In the forward transition, each household's bond position moves to b_{t+1}^*.

Stage 8: Income Shock

The household draws its next-period labor endowment z_{t+1} from the age-dependent Markov transition:

$$\log z_{t+1} = p \log z + \zeta_{a+1} + \nu, \qquad \nu \sim \mathcal{N}(0, \sigma_\nu^2).$$

Let $\pi_a(z' \mid z)$ denote the induced transition probability. The value function is equation (11f):

$$V_a^{\text{postconsume}}(b, z, h, \ell) = \beta \sum_{z'} \pi_a(z' \mid z)\, V_a^{\text{end}}(b,\ z',\ h,\ \ell).$$

In the forward transition, the distribution at each (b, h, ℓ) is spread across z' values:

$$\lambda^{\text{end}}(b, z', h, \ell) = \sum_{z} \pi_a(z' \mid z)\ \lambda^{\text{postconsume}}(b, z, h, \ell).$$

A.1 Within-Period Household Distribution Transition

Composing the distribution transitions across Stages 2–8 defines the within-period household distribution transition operator. Let T_k denote the transition operator for Stage k, defined in the stage descriptions above. Then the within-period transition from λ^{start} to λ^{end} is:

$$\lambda_t^{\text{end}} = (T_8 \circ T_7 \circ T_6 \circ T_5 \circ T_4 \circ T_3)\big(\lambda_t^{\text{start}}\big)\,. \tag{12}$$

The across-period transition appends the entry of new age-one cohorts, the exit of households who have completed age $a_{\max}$, and the realization of aggregate shocks (Stage 9), yielding $\lambda_{t+1}^{\text{start}}$.

A.2 Market Clearing in the One-Period Model

This subsection makes precise the sense in which the equilibrium prices $q_{\ell t}$ and $\rho_{\ell t}$ are the intersection of a supply curve and a demand curve. Each of Equations (4) and (5), the real estate and rental market clearing conditions, are intuitively the intersection of a "supply curve," under which higher prices are associated with higher quantities supplied, and a "demand curve," under which higher prices are associated with lower quantities demanded. However, a household's quantity demanded depends not only on price, but on their expectations of future prices and climate conditions.

However, consider one period of the model equilibrium, with some equilibrium $q_{\ell t}$ and $\rho_{\ell t}$, and end-of-period $V^{\text{end}}(\cdot; \Gamma)$ determined in recursive stochastic equilibrium. Consider that a model with the same beginning-of-period state Γ, and exogenously-imposed end-of-period *payoffs* V^{end} would feature exactly the same equilibrium prices $q_{\ell t}$ and $\rho_{\ell t}$: each household faces exactly the same problem, and no household would change their behavior, relative to the full model.

Therefore, holding Γ and $V^{\text{end}}(\cdot; \Gamma)$ fixed, there is a demand system which depends only on current prices $(q_{\ell t}, \rho_{\ell t})$. The within-period equilibrium prices can thus be found, conditional on these boundary conditions, as the intersection of a supply system and a demand system.

A.3 Perfect-Foresight Transition

In this section I state the equilibrium conditions of the perfect-foresight transition in full and describe how I compute the transition path.

A.3.1 Equilibrium Definition

A perfect-foresight transition path is a sequence of aggregate states Γ_t, as in Equation (6), and prices $\{q_{\ell t}, \rho_{\ell t}\}_\ell$ such that, in every period t of the path: (i) household policy and value functions solve the recursive household problem (11) given Γ_t; (ii) the beginning-of-period distribution λ_t^{start}

is consistent with $\lambda_{t-1}^{\text{start}}$, Γ_{t-1} and household policy functions; (iii) housing stocks evolve by the law of motion below; and (iv) in every location, the ownership market satisfies the complementarity condition

$$H_{\ell t}^{D} \leq H_{\ell t}^{S}, \qquad q_{\ell t} \geq 0, \qquad \left(H_{\ell t}^{S} - H_{\ell t}^{D}\right) q_{\ell t} = 0,$$

and the rental market clears against the stock households hold,

$$H_{\ell t}^{\text{rent}} = \bar{H}_{\ell t}, \qquad \bar{H}_{\ell t} = \min\left(H_{\ell t}^{D},\, H_{\ell t}^{S}\right).$$

The remainder $H_{\ell t}^{\text{disp}} = H_{\ell t}^{S} - \bar{H}_{\ell t}$ is abandoned, and it can be positive only where $q_{\ell t} = 0$. Where the price is positive, $H_{\ell t}^{D} = H_{\ell t}^{S} = \bar{H}_{\ell t}$ and these conditions reduce to the equalities (4) and (5).

Abandoned units leave the model permanently, so the stock that carries forward is the held stock, and the law of motion is

$$H_{\ell t}^{S} = \max\left(c_{\ell}^{-1}(q_{\ell t}),\ (1-\delta^{H})\,\bar{H}_{\ell,t-1}\right),$$

which is Equation (3) written as a single expression. The housing stock recorded in Γ_t is the held stock.

When damages are high enough that the insurance premium exceeds the net rent in Equation (1), $\delta_{\ell t}^{\text{insurance}} > (1-\chi)\,\rho_{\ell t}$, the house price can fall to zero. Free disposal ensures that prices can never be negative, and the preference shocks over locations ensures that the rent will never be zero.

A.3.2 Solution

I first solve the initial steady state and a terminal steady state with GSAT held at its scenario-dependent end-of-horizon value. Given a guess for the path of prices $\{q_{\ell t}, \rho_{\ell t}\}_{\ell,t}$ in the intervening periods, I solve the household problem backward from the terminal steady state's value functions, and I simulate the household distribution and the held housing stock forward from the initial steady state. The excess demands at each date then update the price path, and I iterate until the largest market-clearing residual across all locations, periods and both markets falls below a tolerance. The update is an accelerated fixed-point iteration on the stacked sequence of prices, described in Appendix D.

A.4 Stochastic Transition

The recursive stochastic equilibrium consists of household value and policy functions, an aggregate transition function $\Gamma'(\Gamma, s')$, and price functions $q_{\ell}(\Gamma)$ and $\rho_{\ell}(\Gamma)$.

The climate block of the aggregate state is $D = (t, s, \text{GSAT})$. The scenario s follows the Markov process (7) with a time-invariant transition matrix, and temperature $\text{GSAT}(t, s)$ depends on both year and scenario.

The household distribution evolves by the within-period operator (12), followed by the entry

of new age-one cohorts and the exit of households completing age $a_{\max}$. Potential housing supply follows the law of motion of Appendix A.3,

$$H_\ell^S = \max\big(c_\ell^{-1}(q_\ell(\Gamma)),\ (1-\delta^H)\,H_\ell\big),$$

where H_ℓ is the held-over stock included in Γ. The stock carried into Γ' is $\min(H_\ell^D, H_\ell^S)$, and the lagged price carried into Γ' is $q_\ell(\Gamma)$. The scenario is drawn $s' \sim \mathcal{P}(\cdot \mid s)$, and $\mathrm{GSAT}' = \mathrm{GSAT}(t+1, s')$.

The continuation value (8) takes the expectation over the beginning-of-next-period value, weighted by the transition probabilities $\mathcal{P}(s' \mid s)$.

The price functions satisfy the ownership complementarity condition and the rental clearing condition of Appendix A.3, at every aggregate state Γ and every location ℓ.

B Data

There are three types of data inputs to this model: data describing the 2020 baseline economy, used in Section 3; data mapping global temperature into local fundamentals, used in Section 4; and data informing the stochastic global climate process, used in Section 5.

B.1 Data Inventory

Table B1 lists all data sources. The model's 2447 locations are the 2020 PUMAs of the contiguous United States.

Table B1 gives the sample restrictions of each microdata moment. Earnings are household wage income, summed over household members. The decadal moving share is the share of households whose head moved into the dwelling within the previous ten years. Not-applicable codes of total and wage income are set to zero before PUMA means are taken. Entrant net worth is the quartiles of the 2022 SCF net-worth distribution, scaled down by the ratio of mean net worth among households with heads under 35 to the overall mean. The age–earnings profile is measured in six ten-year bins of the head's age.

A PUMA's 2020 house price is the mean over its counties with a Zillow value, with weights rescaled to sum to one. House values and rents are in 2020 dollars per average home, and the price-to-rent target divides the 2020 price by a decade's rent.

Distances between locations, which are multiplied by F^d to obtain the variable component of bilateral moving costs, are great-circle distances between PUMA centroids, in kilometers. The flow-weighted median distance of moves longer than 20 kilometers is computed from the Census county-to-county migration flows for 2016–2020 (U.S. Census Bureau 2020b) and the NBER county distances (National Bureau of Economic Research 2010), converted from miles.

Table B1—Empirical inputs to the quantification.

Input	Source	Vintage	Spatial unit	Use	Robustness
Panel A. Per-location inputs					
Household microdata	IPUMS USA, ACS household extract	ACS 2020	2020 PUMA	Section 3: The bedroom stock H_ℓ, households, rent and tenure	Occupied units; measured bedrooms, no size adjustment
Person microdata	IPUMS USA, ACS person extract	ACS 2020	2020 PUMA	Section 3: Working-age earnings, the $\bar{A}_\ell$ target	Heads 20–65 on household weights
House-price index	Zillow Home Value Index, county	2020; release not pinned	County to PUMA	Section 3: The 2020 price q_ℓ, and χ's numerator	A typical-home index, not a transaction mean
Housing-supply elasticities[a]	Baum-Snow and Han (2024), finite-mixture instrumental variables	2000 tracts	Tract to PUMA	Section 3: β_ℓ and the supply shifter Π_ℓ	
Flood average annual loss[a]	Bates et al. (2021)	2020 level; 2050 RCP4.5	County to PUMA	Section 4: $\bar{\delta}_\ell$ and its gradient δ^g_ℓ	
Degree-days[a]	NASA NEX-GDDP CMIP5 rasters	CMIP5; RCP4.5 and 8.5, 1985–2095	Raster to PUMA	Section 4: The gradients composed into A^g_ℓ, α^g_ℓ	
Panel B. Global climate					
Global temperature index	IPCC AR5, CMIP5 ensemble	CMIP5 and AR5; 2000–2090	Global	Section 4: The warming axis the gradients are fitted on	A global-mean surface air temperature
Long-horizon projections	Lyon et al. (2022)	2022; RCP4.5 and 6.0 to 2500	Global	Sections 4 and 5: The divergence profile; δ^g's normalizer	Only the shape is data-informed; the rest is swept
Damage sensitivities[a]	Deryugina & Hsiang 2014 (w20750); Albouy et al. (2016)	As published	National	Section 4: The per-degree-day elasticities	
Panel C. National moments					
Price-to-rent ratio	Zillow index over ACS in-place gross rent	2020	National	Section 3: χ's target	Per average home; band [1.47, 1.53]
Rent-to-earnings ratio	IPUMS USA household and person	ACS 2022	National	Section 3: γ's target, given σ	Cash renters, head 20–59, with wage income

Table B1—Continued

Input	Source	Vintage	Spatial unit	Use	Robustness
Panel C. National moments (continued)					
Decadal moving share	IPUMS USA and the public Census PUMS	ACS 2022	National	Section 3: $\overline{F}$'s target	Heads 20–79, fully measured
Migration distance	Census county-to-county flows, NBER distances	Flows 2016–20; distances 2010	County pair	Section 3: F^d's target	A flow-weighted median of moves over 20 km, not a mean
Household net worth	Federal Reserve SCF Summary Extract	SCF 2022	National	Section 3: Entrant wealth, and the bequest motive's target	Entrant quantiles downscaled to ages 20–29
Wealth at 70–79 over 60–69	SCF Bulletin, mean net worth by age	SCF 2022	National	Section 3: An untargeted moment	Weakly identified by the bequest motive
Financial return	SCF, returns on financial assets	SCF 2001–22	National	Section 3: The risk-free return r^f	Computed from the survey files
Age–earnings profile	IPUMS USA, earnings by head-age decade	ACS 2020	National, six bins	Section 3: The age shifters ζ_a	Backed out through the model's income process
Residential share of flood loss	First Street Foundation (2021)	2021	National	Section 4: The residential share of both flood-loss inputs	Uniform across counties and years; band 0.60–0.63
Panel D. Borrowed estimates					
Goods–housing substitution	Albouy, Ehrlich, and Liu (2016)	2016	National	Section 3: σ, external and held out	Sensitivity band [0.487, 0.691]
Idiosyncratic income process	Kaplan, Mitman, and Violante (2020)	2020	National	Section 3: Persistence, innovation and entrant dispersion	Decadalized from an annual process
Equity requirement, realtor fee	Kaplan, Mitman, and Violante (2020)	2020	National	Section 3: κ and ϕ	The fee is internally calibrated in the source
Panel E. Geography and crosswalks					
PUMA boundaries	IPUMS NHGIS shapefile	2020 definitions	2020 PUMA	Section 3: The 2,447-location universe	Contiguous states only
Pairwise distances	Derived from the PUMA shapefile	2020 geography	PUMA pair	Section 3: The distance entering the moving cost	Centroid to centroid
Crosswalks	Census, NHGIS and IPUMS crosswalk files	2020 file; ACS 2021; 2010-to-2020 PUMA land-area crosswalk not pinned	Tract and county to PUMA	Sections 3 and 4: County and tract inputs onto PUMAs	Exact tract nesting on housing-unit shares

County- and tract-level inputs are mapped to the 2447 locations by housing-unit shares, and the degree-day rasters by ground area.

B.2 CMIP5 Heating/Cooling Degree-Days

Local heating and cooling degree-days are the downscaled climate indicators of Gassert, Cornejo, and Nilson (2021), published by the World Resources Institute and derived from the NASA NEX-GDDP archive (NASA Center for Climate Simulation), which downscales CMIP5 model output by the quantile-mapping method of Thrasher et al. (2012). The indicators give annual heating and cooling degree-days against a base of 18.3 °C (65 °F), in °C-days, on a 0.25° grid, for RCP4.5 and RCP8.5 over nine overlapping windows centred on 2000 through 2080. Each layer holds the 25th, 50th and 75th percentiles across the models of Table B2; I use the median. I crosswalk from grid cells to PUMAs by taking the mean over grid cells intersecting its PUMA, weighted by the ground area of each intersection. Section 4 regresses these on global temperature, GSAT, to obtain the slopes XDD_ℓ^g that are used to set A_ℓ^g and α_ℓ^g.

GSAT is the CMIP5 multi-model ensemble mean of global surface air temperature, as an anomaly in °C (Intergovernmental Panel on Climate Change 2013), and is used in the computation of local climate sensitivities A_ℓ^g and α_ℓ^g.

For the aggregate climate process, the extended RCP pathways of Lyon et al. (2022) are used.

I use CMIP5 rather than CMIP6 for consistency with the imported temperature sensitivities (Appendix B.4), which were estimated using CMIP5 climate inputs in the source studies.

B.3 Flood Average Annual Loss

Flood losses are the average annual loss (AAL) estimates and projections of Bates et al. (2021), to my knowledge the only single source that projects combined losses from coastal, fluvial and pluvial flooding. The data owner, the First Street Foundation, has violated the original Data Availability statement of the paper, and the study results are no longer publicly available. I use the subset of the Bates et al. (2021) data provided in the replication package of Wing et al. (2022). The AAL data are county-level, for 2020 and for 2050 under RCP4.5.

The county-level data is crosswalked to PUMAs in proportion, weighted by number of housing units in each PUMA-county intersection, computed from tract-level counts in ACS table B25001 (U.S. Census Bureau 2021), with exact tract-PUMA nesting assignment reported in (U.S. Census Bureau 2020a). The county losses cover all structures, while the model has only housing structures, so I deflate the AAL data by 0.655, the national residential flood loss estimated by First Street Foundation (2021) divided by the total of the county file, uniformly across counties and years.

The 2020 AAL estimates are used as the baseline damage rate $\bar{\delta}_\ell$. The 2020-to-2050 change, divided by the warming of the model's own RCP4.5 pathway over the same years, is used as the gradient δ_ℓ^g:

$$\delta_{\ell t} = \max\left(\bar{\delta}_\ell + \delta_\ell^g \,\mathrm{GSAT}_t,\ 0\right).$$

The model and data align exactly in two places, therefore: the 2020 baseline and the RCP4.5 scenario in 2050. Appendix C gives the remaining arithmetic.

B.4 Imported Climate Sensitivities

The temperature sensitivities of productivity and of amenities are imported from Deryugina and S. M. Hsiang (2014) and Albouy et al. (2016), respectively. Appendix C recenters each from its source's reference temperature and functional form onto the model's heating and cooling degree-days. The sensitivity of productivity to cold is the least certain, because the cold arm of the source estimate is noisy; I use a conservative constant, and Appendix C gives its plausible range. The amenity sensitivity is a compensating differential, and includes any factor correlated with heat, including wildfire risk. That is, the amenity sensitivity captures the extent to which wildfire risk is correlated with temperature in the cross section. The model has no separate wildfire term.

B.5 Housing-Supply Elasticities

Housing-supply elasticities θ_ℓ are taken from the finite-mixture instrumental-variable estimates of Baum-Snow and Han (2024) on 2000 census tracts. I crosswalk them to 2020 PUMAs via 2020 census tracts, as in B.3. Spatial coverage of the Baum-Snow and Han (2024) estimates is incomplete, with coverage caps where density is low. I impute missing values via OLS regression of measured elasticities on log population density, estimated over PUMAs below a density cutoff, censored below at 0.05.

B.6 Acknowledgements and Terms of Use

Several of the providers above attach mandated acknowledgement or attribution wording to the use of their data, and several attach terms that bear on what a replication archive may contain. This subsection discharges the first and records the second.

B.6.1 Mandated Acknowledgements

The heating and cooling degree-days are the PREP downscaled climate indicators of Gassert, Cornejo, and Nilson (2021), published by the World Resources Institute under a Creative Commons Attribution 4.0 International licence and derived from NASA's NEX-GDDP archive.

Climate scenarios used were from the NEX-GDDP dataset, prepared by the Climate Analytics Group and NASA Ames Research Center using the NASA Earth Exchange, and distributed by the NASA Center for Climate Simulation (NCCS) (NASA Center for Climate Simulation). NASA names Thrasher et al. (2012) as the primary reference for the quantile-mapping method behind the archive.

The degree-day rasters and the global warming index both rest on the CMIP5 multi-model archive. We acknowledge the World Climate Research Programme's Working Group on Coupled Modelling, which is responsible for CMIP, and we thank the climate modeling groups (listed in Table B2 of this paper) for producing and making available their model output. For CMIP the U.S. Department of Energy's Program for Climate Model Diagnosis and Intercomparison provides

Table B2—CMIP5 models used in the degree-day projections.

Modeling center	Institute ID	Models
Beijing Climate Center, China Meteorological Administration	BCC	bcc-csm1-1
Canadian Centre for Climate Modelling and Analysis	CCCma	CanESM2
Centre National de Recherches Météorologiques / Centre Européen de Recherche et Formation Avancée en Calcul Scientifique	CNRM-CERFACS	CNRM-CM5
Commonwealth Scientific and Industrial Research Organisation and Bureau of Meteorology, Australia	CSIRO-BOM	ACCESS1-0
Commonwealth Scientific and Industrial Research Organisation in collaboration with the Queensland Climate Change Centre of Excellence	CSIRO-QCCCE	CSIRO-Mk3-6-0
College of Global Change and Earth System Science, Beijing Normal University	GCESS	BNU-ESM
Institute for Numerical Mathematics	INM	inmcm4
Institut Pierre-Simon Laplace	IPSL	IPSL-CM5A-LR, IPSL-CM5A-MR
Atmosphere and Ocean Research Institute (The University of Tokyo), National Institute for Environmental Studies, and Japan Agency for Marine-Earth Science and Technology	MIROC	MIROC5, MIROC-ESM, MIROC-ESM-CHEM
Max Planck Institute for Meteorology	MPI-M	MPI-ESM-LR, MPI-ESM-MR
Meteorological Research Institute	MRI	MRI-CGCM3
National Center for Atmospheric Research	NCAR	CCSM4
Norwegian Climate Centre	NCC	NorESM1-M
Geophysical Fluid Dynamics Laboratory	NOAA GFDL	GFDL-CM3, GFDL-ESM2G, GFDL-ESM2M
National Science Foundation, Department of Energy, National Center for Atmospheric Research	NSF-DOE-NCAR	CESM1-BGC

This is the set of models summarized by the NASA NEX-GDDP degree-day ensemble (Thrasher et al. 2012; Gassert, Cornejo, and Nilson 2021).

coordinating support and led development of software infrastructure in partnership with the Global Organization for Earth System Science Portals.

The household and person microdata are IPUMS USA (IPUMS USA) and the geography is IPUMS NHGIS (Manson et al. 2021). Both are harmonized from source data supplied by the United States Census Bureau, which IPUMS asks be acknowledged as the source of the underlying data. The county house-price index is the Zillow Home Value Index (Zillow); data provided by Zillow Group. The disaster-loss comparison of Appendix C uses SHELDUS (Center for Emergency Management and Homeland Security), whose provider mandates a version-pinned citation naming the database version and its release year.

B.6.2 Terms of Use Bearing on Replication

Four inputs cannot be redistributed with a replication package. The IPUMS terms of use forbid redistribution of the microdata without permission, allowing only the subset needed to meet a journal's data-availability requirement; the extract definitions in the archive therefore stand in place of the microdata themselves. The flood average-annual-loss surface of Bates et al. (2021) is

no longer distributed publicly by its institutional owner, and the county file used here is the subset deposited with the replication materials of Wing et al. (2022), which is where a replicator obtains it. SHELDUS is distributed under an end-user licence agreement that restricts onward sharing, and the extract behind the disaster-loss comparison is not in the archive. The CMIP5 terms of use permit sharing with close collaborators but direct wider redistribution to the ESGF portals rather than to a third-party archive.

Everything else the quantification consumes is redistributable. The Census products — the ACS public-use microdata (U.S. Census Bureau 2022), table B25001 (U.S. Census Bureau 2021), the tract-to-PUMA relationship file (U.S. Census Bureau 2020a) and the county-to-county migration flows (U.S. Census Bureau 2020b) — are United States government works in the public domain, as is the Survey of Consumer Finances (Board of Governors of the Federal Reserve System 2023). The NBER county distance database (National Bureau of Economic Research 2010) is openly distributed. Zillow publishes no redistribution restriction on the research files and asks only for clear attribution.

C Climate Damage Harmonization

This section derives the sensitivities A_ℓ^g, α_ℓ^g and δ_ℓ^g from the imported estimates described in Section 4.

C.1 From Global Temperature to Degree Days

For each location, I estimate equation (9) by ordinary least squares, separately for heating and cooling degree-days, on 18 observations: RCP4.5 and RCP8.5 by nine 30-year periods centred on 2000 through 2080. GSAT is the CMIP5 multi-model ensemble mean of the global-mean surface air temperature anomaly, and the slopes HDD_ℓ^g and CDD_ℓ^g are in °C-days per °C. Table C1 reports their fit.

The slopes, mapped with the 2020 levels in Figure C1, are the only channel through which global temperature affects local productivity and amenities: both temperature sensitivities are weighted sums of them.

C.2 Labor Productivity

From Deryugina and S. M. Hsiang (2014) I take two constants: the effects on log income of an additional heating degree-day, $A^{g,\mathrm{HDD}}$, and of an additional cooling degree-day, $A^{g,\mathrm{CDD}}$. Each is the slope of a chord of their estimated temperature profile, from the degree-day base to a temperature on the cold or the hot side, in log points per °C-day. Both are negative. Because the constants are the same in every location, the composition of Section 4.3 holds exactly, and all variation of A_ℓ^g across locations comes from HDD_ℓ^g and CDD_ℓ^g.

The HDD coefficient on productivity (i.e. the marginal effect of daily temperature on cold days) is the least well determined of the four coefficients described in this section. I use $A^{g,\mathrm{HDD}} =$

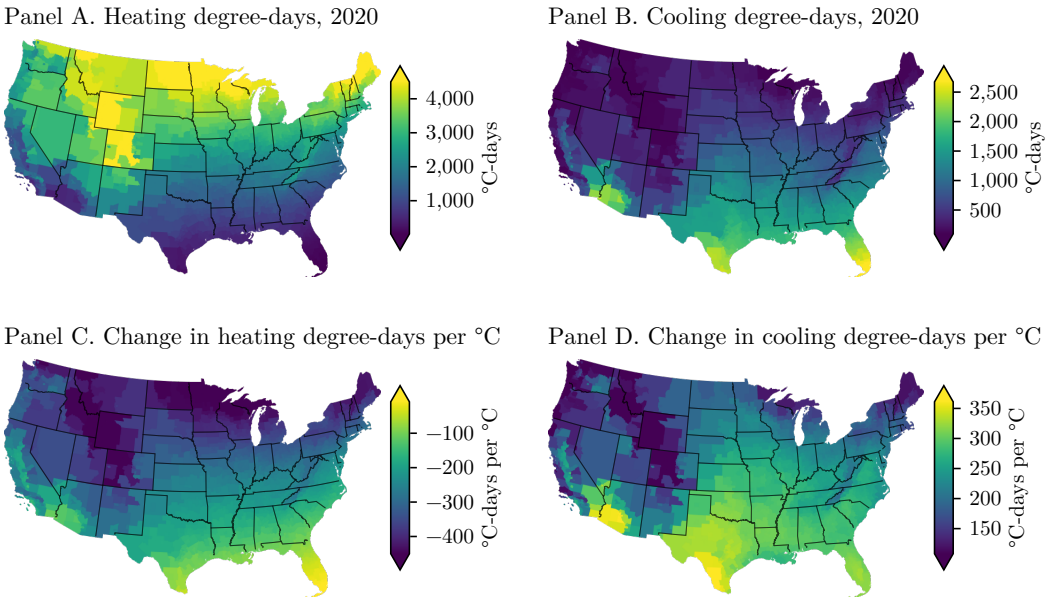


Figure C1. Heating and cooling degree-days by location: levels in 2020 (Panels A and B) and changes per °C of GSAT, HDD_ℓ^g and CDD_ℓ^g (Panels C and D).

Slopes are estimated as in Section 4.1. Scales are clipped at 1st and 99th percentiles.

-8.1×10^{-6}; a plausible range for its magnitude is 0 to 1.8×10^{-5}. Appendix B discusses the source.

The degree-day slopes are per degree of the CMIP5 index. The model's climate state are anchored to Lyon et al. (2022), which warms at a slightly different rate. I convert both temperature sensitivities to the model's index by the ratio of the two indices' warming from 2020 to 2050 under RCP4.5,

$$A_\ell^g = \frac{\Delta\text{GSAT}^{\text{CMIP5}}}{\Delta\text{GSAT}} \left(A^{g,\text{HDD}} \,\text{HDD}_\ell^g + A^{g,\text{CDD}} \,\text{CDD}_\ell^g \right),$$

and similarly for α_ℓ^g, where $\Delta\text{GSAT}^{\text{CMIP5}}$ and ΔGSAT are the same warming on the CMIP5 index and on the model's. A_ℓ^g is then in log points of productivity per degree of the model's index and multiplies the GSAT_t paths directly.

C.3 Disaster Projections

The only disaster channel in the model is combined coastal, fluvial, and pluvial flooding (Appendix B.3). Of the eighteen FEMA hazard categories, earthquake and volcano are not driven by climate. Heat wave, cold wave, winter weather, ice storm, wildfire, drought and hail are correlated with temperature and cannot be separated from the amenity and productivity estimates the model imports. The data source is Center for Emergency Management and Homeland Security, which attributes damages from events with any nonzero wind and flood components evenly to both wind and flooding, which may greatly overstate wind damages and double-count flooding damages if so. The model omits the wind damage of hurricanes, tornado, strong wind, and lightning.

C.4 Heat and Cold Amenities

Albouy et al. (2016) estimate households' willingness to pay to avoid hot days from compensating differentials in real wages across space: interpreting wage residuals and amenities and regressing those residuals on local temperature. Therefore, factors correlated with temperature will load into this estimate to the extent of that correlation, including wildfire risk. That is, part of the burden of wildfire is therefore already in α_ℓ^g, and a separate wildfire term would double count it. The model's treatment of wildfire is nevertheless incomplete, by an amount I cannot observe. The productivity estimates of Deryugina and S. Hsiang (2017), by contrast, are estimated using time series variation in panel data with county fixed effects — controlling for exactly the time-invariant local characteristics that Albouy et al. (2016) use. The latter, similarly, control for wages by design. The two approaches are therefore unlikely to double-count by design.

As with productivity, the constants $\alpha^{g,\mathrm{HDD}}$ and $\alpha^{g,\mathrm{CDD}}$ are chord slopes of the estimated profile, in log points of amenity value per °C-day, and α_ℓ^g is composed from the same degree-day slopes with the same change of origin and of index (Section C.2).

Warming lowers heating degree-days and raises cooling degree-days, so it raises amenities in cold places and lowers them in hot ones.

C.5 Linearity of the Composed Climate Sensitivities

Because the imported local climate sensitivities enter linearly into the composition, each parameter is log-linear in degree-days by construction. Figures 2 and C2 report the degree to which degree-days are indeed linear in global temperature.

Of the 2447 locations, all but 309 have an amenity R^2 above 0.5 and all but 16 a productivity R^2 above 0.5. The median R^2 is 0.9612 and 0.993 respectively. Among the poorly fitted locations the median magnitude of the amenity sensitivity is 0.1103 percent per °C, against 0.9398 among the rest; for productivity the two are 0.05245 and 0.8931. A small sensitivity among the poorly fitted locations means the heating and cooling terms nearly cancel there and the net effect is close to zero.

Table C1 reports the results of these regressions.

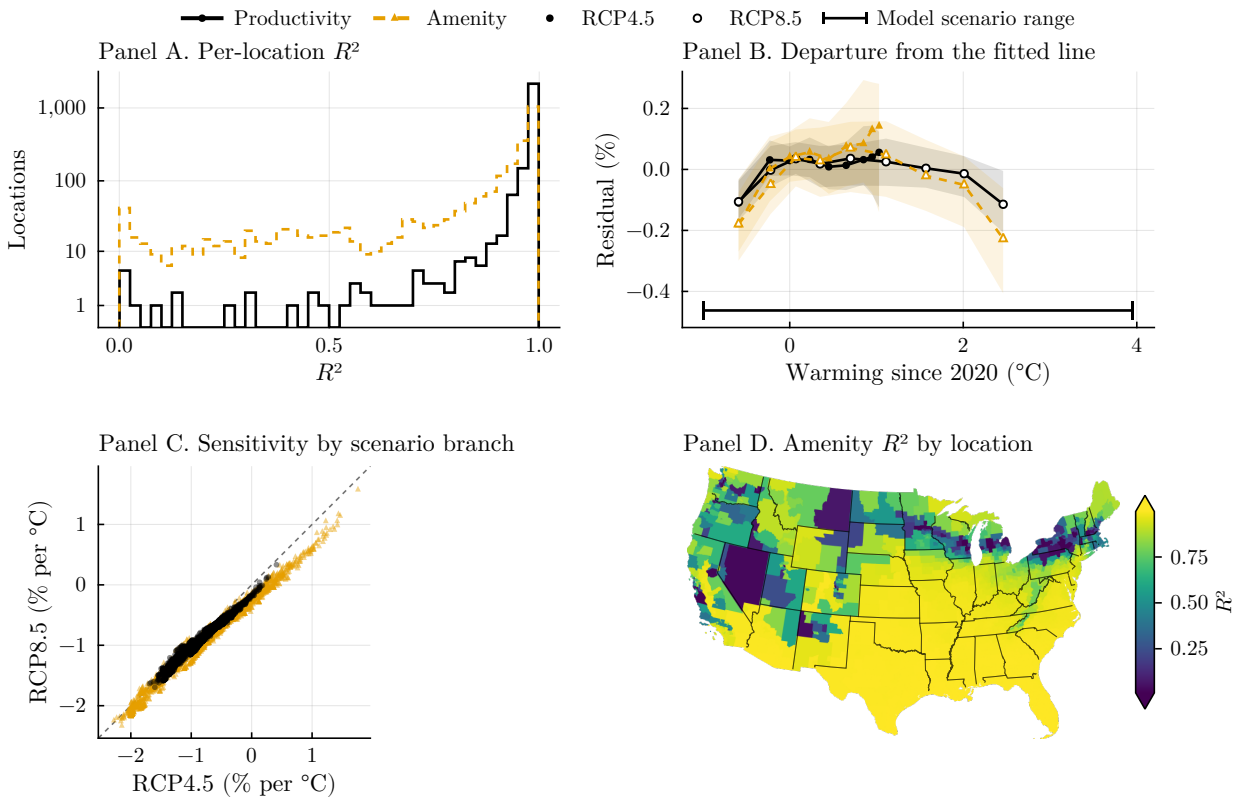


Figure C2. Linearity of log productivity and amenity fundamentals in global-mean surface air temperature, by location. Locations are 2020 PUMAs, numbering 2447 in the contiguous U.S.. Each fit uses 18 observations, two RCP scenarios by nine 30-year periods centred on 2000 through 2080, on the CMIP5 ensemble mean temperature measured from 2020. Panel B plots the median and the 5th–95th percentile band of residuals across locations; the shaded range is the warming the model's scenario nodes span to 2100, on the model's own temperature index. Panel C compares each location's sensitivity estimated on each scenario alone. The scale in Panel D is clipped at the 1st and 99th percentiles.

Table C1—Location-by-location regressions of degree-days and log fundamentals on global-mean surface air temperature. Each row summarizes 2447 regressions, one per PUMA. Each has 18 observations comprising two RCP scenarios by nine 30-year periods centred on decades 2000 through 2080. The regressor is CMIP5 ensemble mean temperature anomaly. The pooled R^2 sums squared residuals and squared deviations from location means over all locations. Panel A reports outcome variables HDD_ℓ^g and CDD_ℓ^g. Panel B reports outcome variables log fundamentals, derived from degree-days as in Section 4.3.

	Fitted slope across locations			Fit		
Regression	5th percentile	Median	95th percentile	Median R^2	Pooled R^2	Share of locations with R^2 below 0.5 (percent)
Panel A. Degree-days on global temperature (°C-days per °C)						
Heating degree-days	−393.7	−251.4	−50.1	0.990	0.991	0.0
Cooling degree-days	155.3	241.0	327.0	0.996	0.995	0.0
Panel B. Composed log fundamental on global temperature (percent per °C)						
Productivity	−1.43	−0.89	−0.43	0.993	0.993	0.7
Amenity	−1.99	−0.75	0.15	0.961	0.979	12.6

D Computational Method

This appendix describes the computational method and solution validation for the stochastic transition under aggregate climate uncertainty. It is a variant of neural Value Function Iteration (nVFI), introduced in full generality in Sun (2026a). In order to efficiently generate training data, a fast household block solver was developed and is introduced in full generality in Sun (2026b).

D.1 Stage Decomposition

Each period of the within-period model is solved by decomposing it into the stages described in Section 2.1.9. Given the end-of-period value V^{end} and a guess of the period's prices, I solve the household problem (11) backward through the stages, simulate the household distribution forward through them, compute market clearing conditions and update prices until both markets clear. In the conventional solver, used for the perfect-foresight solution and described in Appendix A.3, V^{end} is taken by iterating backward from the following period. The global nVFI solver obtains V^{end} as the output of a neural network whose inputs are the beginning-of-period aggregate state.

D.2 Neural-Network Value Function

A neural network with 2456961 parameters approximates V^{end} as a function of end-of-period household state and the beginning-of-period aggregate state Γ_t (6), over a household state space of 66289230 grid points.

The structure of the neural network consists of three parts. The first part uses three encoders to summarize the state at each location: one which encodes location-level information not including the household state distribution, one which encodes the household state distribution, and one which integrates the two summaries across space to obtain an economy-wide summary. The first encoder takes as input a location's fundamentals $\bar{A}_\ell$, $\bar{\alpha}_\ell$ and $\bar{\delta}_\ell$, their climate sensitivities A^g_ℓ, α^g_ℓ and δ^g_ℓ, its supply parameters Π_ℓ and θ_ℓ, the previous period's housing stock and house price, and a location indicator. A second encoder maps each point of the household state grid to a vector, which is integrated over the location's household distribution λ_t^{start} to summarize the household state distribution. This is a spatial extension of the "Generalized Moments" architecture of Han, Yang, and E (2021). The full summary of a location is obtained by combining these summaries via a FiLM layer. A third encoder maps each location's summary to a vector, and the average over locations summarizes the cross-section.

The second part summarizes the climate: global temperature GSAT_t, the mean and standard deviation of temperature in each of the next three periods, the decade, and the current climate scenario index. The third part maps the household's own state, with its location's summary and the cross-sectional and climate summaries, to the value. The network takes the current climate scenario index as an input.

D.3 Training

I train the network by neural value function iteration (nVFI). It is pre-trained using the end-of-period values along the perfect-foresight median path, then further pre-trained using true nVFI but on a variant of the full model with the variance of the aggregate climate process set to zero.

Each epoch of nVFI simulates the transition forward using the current network along a climate path which is not foreseen in advance. The training labels are obtained by one-period lookahead along all possible draws of the one-period-ahead climate scenario, with the expectation taken weighted by the true transition probabilities.

D.4 Software

The model is solved and the network trained in Julia (Bezanson et al. 2017), on the GPU through CUDA.jl (Besard, Foket, and De Sutter 2019; Besard et al. 2019). The value network is built and trained with Flux.jl (Innes 2018b; Innes et al. 2018), whose gradients come from the reverse-mode automatic differentiation of Zygote.jl (Innes 2018a); the stochastic climate state is drawn through Distributions.jl (Besançon et al. 2021). Tabular work runs through DataFrames.jl (Bouchet-Valat and Kamiński 2023), exhibits are drawn with Makie.jl (Danisch and Krumbiegel 2021) and its Cairo, GL and Geo backends, and map geometry is simplified with GeometryOps.jl (Singhvi, Arnott, and GeometryOps contributors 2023). Alongside these the pipeline uses CSV.jl for tabular input and output, JLD2.jl for network checkpoints, StatsBase.jl, GLM.jl and FixedEffectModels.jl for the calibration moments, and QuantEcon.jl for the Rouwenhorst discretization of the climate autoregression (Rouwenhorst 1995); none of these publishes a citation request.

The data pre-transform runs in Python on NumPy (Harris et al. 2020) and pandas (McKinney 2010; The pandas development team 2020). The spatial construction — carrying county and tract quantities onto the 2447-location partition and weighting the degree-day rasters onto PUMA polygons — runs on GeoPandas (Fleischmann et al. 2026), Shapely (Gillies et al.) and rasterio (Gillies and Rasterio contributors 2013), and through them on the GDAL (GDAL/OGR contributors), PROJ (PROJ contributors) and GEOS (GEOS contributors 2025) libraries, which carry the raster I/O, the coordinate transformations and the geometry operations respectively.

The results were produced on one NVIDIA GeForce RTX 4090 (driver 595.91.07) with Julia 1.12.6, CUDA.jl 5.11.0 and the CUDA 13.2 toolkit.